Democratic Accountability and Environmental Regulation

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August 10, 2026

Abstract

Environmental governance presents a core challenge for democracies: pollution yields concentrated benefits and diffuse harms, weakening incentives for political action. We investigate when politicians act to curb pollution, and what breaks this accountability, in the context of crop-residue burning in South Asia, which kills an estimated 2 million people annually. Linking satellite-detected fires, wind fields, and political data across 798 rural constituencies in northern India over a decade, we use a difference-in-differences design that exploits wind-direction shifts altering whether smoke from a given location drifts into a politician's own constituency or a neighbor's. When their constituents bear the pollution costs, politicians curb fires by 13 percent. Yet two forces systematically undermine this accountability: sanctioning weakens as organized farmer protests pressure politicians to tolerate burning, and selection fails when constituencies elect politicians with private agricultural interests, both increasing fires even where politicians' own voters are harmed. These findings demonstrate that democratic accountability can generate environmental protection under diffuse incentives, but sustaining it requires institutions robust to the concentrated interests that benefit from pollution.

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1 Introduction

Few areas of governance reveal the limits of democratic accountability more clearly than environmental management. A growing body of evidence shows that voters rarely prioritize environmental issues (?), leaving politicians with little electoral incentive to address pollution. Even when citizens do care about environmental problems, these preferences seldom translate into policies addressing environmental damage (??).[1] Instead, environmental regulation often reflects the influence of concentrated and organized lobbies (????), reinforcing a growing pessimism about the prospects for democratic political action against environmental degradation.

Yet this pessimism raises a crucial question: do politicians ever act to protect the environment, and if so, under what conditions?[2] Studying this question is challenging because environmental degradation, unlike most policy domains, is a slow-moving process whose harms unfold gradually and often escape citizens' immediate perception. Environmental humanities scholars have described this temporal disjuncture as "slow violence" (?), and political scientists argue that this very disconnect between environmental harm and lived experience helps explain why environmental issues remain politically marginal and why democratic systems so often struggle to generate sustained action against them (?).

In contrast to the powerful incentives for inaction on the environment described above, we instead document that there is still a promise of accountability: we show that politicians do take meaningful steps to curb pollution when their own constituents bear its toxic effects. However, we also demonstrate that this control is systematically undermined. Building on classic literature, we examine two key channels of political accountability whose erosion can break down politicians' responsiveness to citizens' welfare: sanctioning and selection. Specifically, we highlight how sanctioning weakens under pressure from multiple principals, as citizens' interests compete with those of concentrated lobbies, and how political selection falters when politicians' self-interest diverges from citizens' well-being (?). Understanding these mechanisms is crucial for designing institutions that can channel democratic accountability towards durable environmental improvements.

We study these dynamics in the context of South Asia's air pollution crisis, one of the largest public health emergencies on the planet estimated to kill 2 million people every year, with children as one of its most vulnerable victims (?).[3] Each winter, a thick smog so vast it is visible from space, blankets northern India as millions of farmers burn leftover crop residue after rice harvest. Roughly 60 percent of this seasonal pollution originates from these fires (??). Figure 1 shows Delhi during smog season. Despite the scale of the crisis, the visibility and consequentiality of the problem, and the illegality of crop burning, the practice continues today. This paper studies how politics explains variation in this persistence.

The first part of the paper examines how democratic accountability shapes political action towards environmental regulation. We propose that politicians will be most incentivized

[1] Examples of successful action on pollution come from China (see ? for a review), where, however, politicians are unelected and face different incentives than in democracies (?). Moreover, scholars increasingly document that a transition to a democracy may be associated with poorer environmental outcomes (??).

[2] In contrast to the political context, scholars have recently examined these questions within bureaucracies (??).

[3] In comparison, COVID-19 at its peak killed 6 million people in the first two years worldwide.

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Figure 1: Gate of India, Delhi, during and after the smog season (Source: ?)

to stop crop burning in places from which smoke pollutes their own constituents, rather than a neighbors'. We leverage wind as a source of variation to resolve the challenge of identifying when politicians' incentives bind as wind direction determines where smoke from a burning field would travel. Because seasonal wind patterns in the region are broadly predictable, politicians will have reasonable expectations about them. When wind direction shifts in a month, the same 5km^2 grid-cell in a politician's constituency will switch from sending smoke into neighboring constituencies to sending it into the politician's own, raising the political cost of inaction. We exploit these within-location changes in a difference-in-differences design, combining satellite-detected fires from VIIRS and MODIS (?), and wind direction and speed from ? across 798 rural political jurisdictions in northern India over ten years (2012–2022), for approximately 9 million observations.

We find that fires decrease by 13 percent right after a change in wind direction switches a location from polluting neighboring jurisdictions to a politician's own constituency, an effect that is 34 percent larger in rice producing areas. These results are robust to a host of tests that relax the identifying assumption of the DiD design and vary how we define the outcome, the treatment, and the sample.

We further bolster our interpretation by ruling out that these findings are not explained by the incentives of two key non-politician actors: farmers and bureaucrats. First, if the decline were driven by farmers' pro-sociality rather than by political accountability, we should observe fewer fires when more people live downwind from farmers. In fact, the opposite holds: farmers tend to burn more when their crop burning is likely to affect more people, and the reduction in fires emerges only when political incentives are active. Second, to test if bureaucrats' incentives explain our results instead of politicians', we use the fact that the jurisdictions for the two actors overlap imperfectly in India, allowing us to examine the impact of the same location being a source of pollution for a politician's constituents, the bureaucrat's constituents, both, or neither. We show that our results persist when a location is a source of pollution for politicians but not for bureaucrats, confirming that the logic of political incentives drives fires.[4]

[4] Interestingly, we also find that when the location pollutes both the politician's and bureaucrat's jurisdictions, the total reduction in fires is lower than when the location is a source of pollution only for politicians or only for bureaucrats. This is consistent with the interpretation that political effort is sensitive to free-riding considerations in the presence of incentives for action for other government agents.

Ultimately, this first finding provides powerful evidence that there are prospects for political accountability in environmental management even in a setting where action would seem unlikely: indeed, our evidence suggests that politicians are sensitive to environmental pollution when the well-being of their own constituents is at stake. We use this accountability effect as a benchmark to study its limits, focusing on the channels of weakening sanctions via multiple principals, farmers and citizens, and adverse selection through the election of self-interested politicians.

The second part of the paper examines how accountability weakens when it competes with the interests of actors who benefit from pollution-intensive activities: farmers organized into lobbying groups, and politicians with agricultural ties. Around the world, from truck driver strikes in South Korea, Canada, and the UK, to teacher protests during Covid, sectors have lobbied for state intervention amid rising economic pressures. We investigate this dynamic in the context of the largest protest movement in the world's history, the Farmers' Movement of 2020–21 in India, during which more than 25 million farmers mobilized to secure policy concessions, including the decriminalization of crop burning. We examine the impacts of 390 farmers' protest events (2020-2021) on subsequent crop burning in a stacked difference-in-differences framework.

We show that crop-residue fires increase by 21 percent following protests, consistent with politicians relaxing environmental control in response to intense farmer lobbying. Having demonstrated that protests increased the production of pollution, we next show that they also erode the accountability incentives to politicians' own constituents that we document in the first part of this paper. To do so, we compare the reduction in fires produced by accountability effects before and after protests, and show that in places that did not have protests, the accountability-driven decrease in fires persists. In contrast, places that switch to having protests show significantly more crop burning afterwards. This suggests that politicians relax environmental control in response to concentrated lobbying pressure, even when such inaction harms their own constituents.

In addition to lobbying, we examine another set of actors who may benefit from pollution: politicians with agricultural ties. To capture these interests, we use variation in politicians' professional involvement in agriculture, which may give them a direct economic stake in practices associated with crop burning. We proxy these ties by collecting the candidacy papers of all 2,565 elected politicians in our ten-year sample and coding whether they are agriculturalists by profession. We then estimate the impact of a constituency switching from a non-agriculturalist to an agriculturalist politician on crop burning using a stacked difference-in-differences framework.

We find that when a constituency switches to electing an agriculturalist politician, the rate of crop burning increases by 24 percent, indicating a more permissive policy environment when politicians have personal stakes in agriculture. We next test whether such self-interest also weakens the accountability mechanisms documented in the first part of the paper. Specifically, we compare the reduction in fires driven by accountability before and after a constituency switches to electing an agricultural politician. When non-agricultural politicians remain in office, fires decline when politicians' own constituents are polluted, as expected. In contrast, after constituencies switch to agricultural politicians, crop burning increases even where accountability incentives to reduce it are present. This pattern indicates that, as with lobbying, self-interest can override the pressure of political accountability, even when citizens directly experience the consequences.

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Contributions: This paper makes several contributions. First, we contribute to the literature on how governments regulate the environment, where previous research has tended to focuse on the bureaucracy.[5] We expand this literature by centering the role of politicians: we ask whether and under what conditions politicians in democracies translate citizen interests into better environmental regulation. This contrasts with most studies that portray politicians as obstacles to, not agents of, environmental protection.[6] In doing so, we join very recent work that has begun to probe these questions in the context of political selection by asking if certain politician identity, such as gender and ethnic minorities, improves environmental governance (**??????**). In contrast to the role of identity, we focus on accountability incentives and document its promise as one of the few levers that can improve environmental governance.

The second strand of literature we contribute to is on the politics of air pollution. Despite its scale and urgency, air pollution has received little systematic attention in political science.[7] This deficit is striking because transboundary air pollution presents a particularly poignant challenge to democratic accountability: **?** show that polluting plants are more likely to be located at US state boundaries to potentially offset localized pollution impacts, while **?** show that international Chinese air pollution shapes blame attribution in South Korea.[8] Probing if regulatory accountability can be leveraged to control air pollution, scholars have thus far tended to focus on the demand side, examining conditions under which citizens consider air pollution to be a political problem. Instead, we turn attention towards the supply side, showing that politicians are sensitive to transboundary incentives, take action to prevent harm to their own constituents, yet are swayed by weakening sanctions and selection. By doing so, we bring the study of air pollution into the core of political economy and contribute to a broader understanding of when and why politicians act to confront diffuse environmental harms.

Finally, we also make a contribution to the study of environmental politics in South Asia. Nowhere is the problem of air pollution more severe than in South Asia, home to one in four humans, which now bears the world's highest levels of ambient particulate pollution (**?**). Scholars across disciplines have started investigating the causes and consequences of crop burning, one of the largest contributors to this crisis.[9] We provide the first systematic, large-scale analysis of the political accountability incentives that drive both the persistence and reduction of crop burning in Northern India, the region where most of the subcontinent's air pollution originates and where political action carries profound implications for the lives of millions.

[5] See, for instance, **????**.

[6] For instance, scholars have shown that campaign donations and lobbying can depress pro-environment votes and legal enforcement (**????**), elections can exacerbate environmental harms (**???**) (but c.f. **?**), political control of bureaucratic assignments weakens implementation (**?**), and parties in power can offload conservation costs onto their opponents (**?**). Even informational nudges about voter preferences often fail to lead politicians to adopt pro-environmental policies (**?**).

[7] See **?** and **?** for recent reviews on environment and climate change more broadly

[8] A related literature in economics also studies transboundary pollution in the context of industrial emissions (**???**).

[9] For a selection, see: **??????**

2 The Management of Air Pollution

2.1 Crop Burning in South Asia

Air pollution is one of the foremost threats to human capital and development worldwide. Exposure to fine particulate matter (PM2.5) causes strokes, chronic respiratory disease, diabetes, and premature mortality, with cascading effects on labor productivity, educational attainment, and hospitalization rates (**???**). The global economic cost of air pollution is estimated at nearly $2.9 trillion annually, or roughly 3.3 percent of global GDP (**?**). This burden falls disproportionately on the developing world: South Asia now bears the highest levels of ambient PM2.5 on the planet (**?**), and recent epidemiological work attributes between 1 and 2 million premature deaths per year in India alone to ambient air pollution (**??**).

Agricultural emissions, and crop-residue burning in particular, are among the largest contributors to seasonal PM2.5 spikes in South Asia (**??**). Crop burning is the practice of setting fire to the stubble left in fields after mechanical harvesting to clear the land quickly and cheaply before the next sowing cycle. As farmers themselves acknowledge, burning is a rational response to tight economic constraints: clearing a field with a matchbox costs virtually nothing compared to the thousands of rupees required to rent a mechanized residue-management alternative (**??**). Yet the aggregate consequences are severe. Each post-monsoon season, millions of fires across Punjab, Haryana, and western Uttar Pradesh generate a vast plume of smoke visible from space, blanketing northern India in smog that persists for weeks (**??**).

The roots of this crisis trace to the Green Revolution. The introduction of high-yielding rice and wheat varieties, combined with public procurement guarantees and subsidized inputs, entrenched a rice-wheat double-cropping system across the Indo-Gangetic Plain that leaves farmers with as little as ten to fifteen days between the rice harvest and wheat sowing (**??**). The Punjab Preservation of Subsoil Water Act of 2009, which delayed paddy transplanting until mid-June to conserve groundwater, compressed this window further, increasing farmers' reliance on fire as the fastest residue-clearing method (**??**).

2.2 Political Salience of Crop Burning

Crop residue burning is increasingly a politically salient environmental issue, creating electoral incentives for politicians to claim credit for reducing burning or to avoid blame when it persists. In a representative survey of more than 2,200 residents across all districts of Punjab conducted in 2020–21, **?** asked respondents, "What do you think about stubble burning in Punjab?" Nearly two-thirds (65.1%) described stubble burning as a "Big problem, and have to stop now." In addition, in a survey of 3,116 Delhi residents, (**?**) find that citizens blame the government for air pollution more than any other actor, and that 75 percent hold the state government of Punjab, where the burning occurs, responsible for solving crop-burning pollution, rather than the government of the jurisdiction where the smoke lands. Together, these two studies indicate broad public concern about air pollution, as well as sophisticated blame attribution.

Gemma have a look at this

In Punjab, state-level political discourse tends to center on the scale of subsidies for CRM machinery, the construction of biomass and bio-CNG plants, and the allocation of budgetary resources to stubble management. For example, the Governor of Punjab's address to the state legislature in 2024 highlighted that the government had provided subsidies of 80 percent to farmer groups and 50 percent to individual farmers since 2018, distributing over 117,000 machines, alongside investments in compressed biogas projects designed to absorb surplus paddy straw.[10] In Haryana's Vidhan Sabha, legislators have highlighted both punitive and incentive-based measures, from per-acre fines imposed by the Pollution Control Board to subsidies for straw management projects and briquette-making units.[11]

Importantly, opposition politicians use the same issue to hold incumbents accountable for implementation failures. In Punjab's Vidhan Sabha debates, opposition members have alleged that funds earmarked for stubble management were diverted, and that promised per-acre support never materialized on the ground.[12] This pattern of credit-claiming by incumbents and blame-attribution by opponents is consistent with a setting in which public opinion creates electoral incentives for responsiveness on environmental governance.

Crop burning appears regularly in wider public discourse as well: journalists, courts, and civil society groups routinely pressure the central and state governments to demonstrate progress, particularly during the winter smog season when air quality indices reach hazardous levels.

Taken together, the institutional architecture of anti-burning policy in India creates a setting in which politicians are important actors. They occupy the implementation node where national mandates meet local realities: they influence the deployment of machinery, the intensity of enforcement, the distribution of subsidies, and the political messaging around environmental governance. This makes the study of their incentives, and the conditions under which those incentives bind or break down, central to understanding the persistence and variation of crop burning across northern India.

2.3 State Politicians' Role in Curbing Crop Burning

The governance of crop burning in India involves multiple tiers of government. At the national level, the Commission for Air Quality Management (CAQM), established as a statutory body in 2021, coordinates anti-pollution frameworks across the National Capital Region and adjoining states, combining in-situ residue management support, such as the subsidized distribution of Happy Seeders, mulchers, and balers, with ex-situ programs to channel crop residue toward industrial uses in power plants and brick kilns.[13] The CAQM directs states to prepare district-level and block-level action plans targeting hotspot villages, and its 2024 amendment rules substantially increased the environmental compensation penalties for violations.[14] Nevertheless, because air pollution crosses jurisdictions, effective control requires not

[10] Address of the Governor of Punjab, March 1, 2024. See also the Punjab Governor's Address of January 2020, which detailed the distribution of 28,609 machines through 5,439 custom hiring centers.

[11] Haryana Vidhan Sabha, Bulletin No. 3, September 11, 2018; and the 45th Report of the Committee on the Welfare of SC/ST/OBC, 2021–22.

[12] Punjab Vidhan Sabha Debates, November 29, 2017.

[13] See the Commission for Air Quality Management in National Capital Region and Adjoining Areas Act, 2021; and **?**.

[14] See CAQM Direction No. 84 (October 10, 2024) and the revised schedule of penalties notified in December 2024.

just higher-level coordination but local-level implementation that is responsive to political incentives (?).

State legislators, Members of the Legislative Assembly (MLAs), are important actors in translating these national frameworks into local enforcement and service delivery. In India's federal system, MLAs elected from single-member assembly constituencies (ACs) exercise considerable influence over district administrations. They shape the allocation and deployment of subsidized crop-residue management (CRM) machinery through custom hiring centers, direct the attention of nodal officers and village-level functionaries toward enforcement and monitoring, and serve as important interlocutors between farmers and the state bureaucracy on agricultural policy. This role is reinforced by the structure of India's anti-burning policy architecture. Since the central government launched the Promotion of Agricultural Mechanisation for In-Situ Management of Crop Residue scheme in 2018, over 100,000 CRM machines have been distributed across Punjab, Haryana, Uttar Pradesh, and Delhi, with subsidies of 50 percent to individual farmers and 80 percent to cooperatives and panchayats.[15] The utilization of this machinery, however, depends on local-level logistics, such as timely delivery, operator training, equitable access across landholding sizes, and the resolution of disputes over service charges, tasks that are fundamentally shaped by politicians' convening power and constituency service incentives (??).

Beyond this informal influence, elected representatives also have a formal institutional role in overseeing scheme implementation at the district level. The District Development Coordination and Monitoring Committee (DISHA), constituted by the Ministry of Rural Development since 2016, brings together Members of Parliament, MLAs, and the district bureaucracy in a structured oversight body.[16] Each district's DISHA committee is chaired by the senior-most Lok Sabha MP, with all MLAs from the district serving as members and the District Collector acting as convenor. Meeting quarterly, the committee reviews the implementation of central government schemes, now covering over 100 programs across 37 ministries, including agricultural mechanization and crop residue management, and has formal powers to seek information, identify bottlenecks, and demand follow-up action from district officials.[17] This institutional arrangement gives MLAs a formal channel, alongside their informal leverage, through which they can press district bureaucracies on CRM machinery utilization, anti-burning enforcement, and alternative residue-management programs, or conversely, signal leniency when facing pressure from farmer constituents.

This combination of formal and informal influence extends to enforcement. District magistrates and pollution control boards impose environmental compensation penalties on farmers who burn residue, ranging from Rs. 2,500 to Rs. 15,000 per incident depending on landholding size.[18] MLAs can intensify or relax the pressure behind this enforcement: they may direct district officials to step up patrols or put pressure on state level bodies to use remote-sensing dashboards to target hotspot villages during peak burning season. Alterna-

[15] Ministry of Agriculture and Farmers Welfare, Government of India. Between 2018-19 and 2025-26, cumulative releases under the scheme totaled approximately Rs. 4,237 crore. See ?PIB press release, October 2024.

[16] DISHA superseded the earlier District Vigilance and Monitoring Committees. See ?.

[17] As of 2024, DISHA chairpersons have been appointed in 689 of 718 districts. See ?.

[18] See the Haryana Vidhan Sabha Bulletin No. 3 (September 11, 2018) for penalty schedules. https://haryanaassembly.gov.in/wp-content/uploads/2019/01/Bulletin-No.-3-119181st-Sitting.pdf

tively, they may signal tolerance for burning in response to farmer grievances. Enforcement priorities can be set at the start of the burning season and adjusted as conditions evolve, making political action responsive on a time-sensitive scale. This discretion makes politicians a critical node in the implementation chain, both potentially enabling and constraining the effectiveness of national policy.

3 Politicians' Incentives to Control Fires

Our first objective is to establish whether accountability incentives drive politicians' action to control environmental damage. In this section, we describe the data, empirical strategy, and results showing that responsiveness to constituents' exposure to pollution shapes politicians' behavior to control crop related fires.

3.1 Data

We assemble data on the identity and characteristics of the politicians in charge of each constituency in the sample, as well as on the behavior of farmers, and spatially merge these with information on fires and wind direction. We begin by describing the political data.

Assembly Constituencies: We focus on the geographical jurisdiction of the State Legislative Assembly constituency, defined according to the 2008 delimitation, which was implemented under the Delimitation Act, 2002 (**?**) and based on the 2001 Census. We focus on the states where crop burning is a major problem —Punjab, Haryana, Uttar Pradesh, and Bihar— restricting the sample to rural areas by dropping urban grids. We classify a grid as urban if it contains at least one pixel labeled "Urban" or "Dense Urban Cluster" according to the GHS-SMOD R2023A dataset (**?**), which classifies $1km^2$ pixels according to population size and density thresholds to capture the full settlement hierarchy. Constituencies that consist entirely of urban grids are dropped from the sample, leaving 798 of the 853 assembly constituencies.

Politicians' Incentives: We proxy politicians' incentives to curb fires using variation in wind direction, which determines whether smoke from fires is likely to affect politicians' own constituency or a neighboring one. Wind direction and speed data come from the **?**, which is precise at a 25x25 km resolution. To approximate what politicians might expect for a given month–year, we compute ten-year rolling averages of wind direction using data from the early 2000s onward. Figure B.1 (right panel) shows that actual wind directions and their rolling averages closely track each other, substantiating the expectations that politicians are aware of prevalent wind directions in a season. In this setting, indeed, the two major seasonal wind shifts occur in June and August each year. Section 3.2 explains in more detail how we define political incentives by combining data on wind direction and constituency shapes.

Fires: Our main outcome of interest is the presence and extent of crop stubble burning. We measure crop burning with data on fires from Nasa satellites Moderate Resolution Imaging Spectroradiometer (MODIS, 1km x 1km resolution) and Visible Infrared Imaging Radiometer

Suite (VIIRS, 375m x 375m resolution) from September 2012, the first year in which VIIRS starts operating, to August 2022. The satellite records fires from thermal signatures, such that the presence of cloud cover or smoke in the atmosphere does not prevent its detection. The absence of forests in the area of India we study means that we can safely attribute fires to crop burning activity rather than to forest fires. We plot the total number of fires over time in Figure B.1 (left panel): each year, there are two peaks in the months of April-May and October-November, in correspondence with the crop burning seasons of wheat and rice respectively (**?**).[19]

Crops: Rice is the crop responsible for the largest share of stubble burning, followed—though to a lesser extent—by wheat. To focus on areas most affected by crop burning, we collect data on average crop area and production in 2010, the latest year with micro-level crop data available before the start of our analysis period. The data is sourced from the **?**, it has a resolution of 5 x 5 arc/min, corresponding to approximately twice the size of a grid in the final database, and is plotted in Figure B.2.

Our full sample includes 76,800 grid-cells which we observe in each month and year from September 2012 to August 2022, for a total of 9,216,000 observations. We map each constituency in the sample with an illustration of the gridded data in Figure 2a.

3.2 Empirical Strategy

Treatment Definition: Our treatment is designed to capture a specific accountability incentive. A politician has stronger reason to prevent crop burning in a location when the resulting smoke is likely to settle over their own constituents rather than blow into a neighboring constituency. Whether a given location does so depends on two things: where the location sits within the constituency, and which way the wind is blowing.

We operationalize this at the grid-cell level. For each 5km^2 grid-cell centroid, we draw a line through the centroid perpendicular to the prevailing wind direction vector for that month. This line partitions the constituency into a downwind area, which would receive smoke from a fire in that grid-cell, and an upwind area, which would not. We then compute the constituency population living in each area.

We define a grid-cell as *treated* in a given month-year if the constituency population downwind of the grid-cell exceeds the constituency population upwind of it: a fire in a treated cell would impose pollution costs on most of the politician's own voters, while a fire in an untreated cell would carry most of its smoke past them.[20]

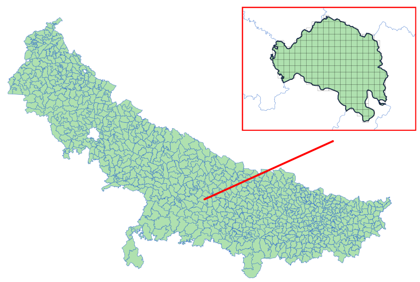
For the prevailing wind direction we use ten-year rolling averages by month rather than realized contemporaneous wind. Seasonal wind in northern India is governed by the monsoon system, with two major directional reversals occurring at approximately known points in the calendar. Politicians can therefore anticipate, before the burning season, which parts of their constituency will bear the pollution costs of a fire in a given month. The monthly rolling average is intended to approximate that expectation, not to measure the wind on any particular day.

[19] Data on fires and wind are taken from **?**.

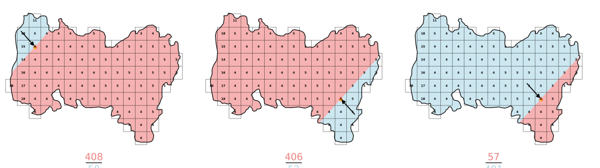
[20] We show robustness to other definitions of treatment that are continuous or otherwise transformed below.

Figure 2: **How Changes in Wind Direction Affect Political Calculus and Crop Burning**

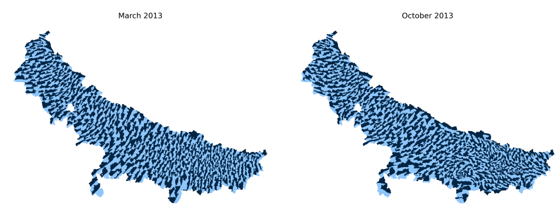
(a) Gridded Dataset



(b) Treatment Definition and Variation Example



(c) Variation in Treatment Over Time Across Units



Notes: *(a)* The map illustrates the 5km × 5km gridded dataset in green with constituency boundaries in blue. It shows the Bachhrawan assembly constituency magnified as an example. *(b)* The figures illustrate the coding of treatment. The blue arrow represents wind direction in a given month-year, and the numbers represent population in a grid. In the left panel, the population in the downwind area from a potential crop burning is larger than the population in the upwind area: we define this grid as treated. In the central panel, the downwind area is larger than the upwind area, but the population in the downwind area is smaller: we define this grid as control. In the right panel, a change in wind direction converts the same grid into the treatment group because a potential crop burning would now pollute the majority of the constituency's population. *(c)* The maps display the spatial distribution of this treatment and its temporal variation in two different months. Treated grids are shaded in dark blue and control grids in light blue. An image covering all periods of analysis is located in this link.

Because wind direction shifts across the calendar, the same grid-cell can be treated in some months and untreated in others: a fire there would pollute its own constituency's voters in one month and a neighbor's in another. This within-grid-cell, across-month switching is the source of our identifying variation.

We present an example of treatment definition and change in status for a grid in our sample in Figure 2b. In Figure 2c we also illustrate how treatment varies in the full sample over two given months.

Figure 2b illustrates the coding rule and shows a grid-cell in our sample changing status as the wind turns. Figure 2c shows the spatial distribution of treatment in two different months.

Estimation Strategy: Using this treatment definition to capture politicians' accountability incentives, we estimate a stacked-by-event difference-in-differences model for the impact of a grid-cell switching to polluting most of its jurisdiction's population on fires, leveraging within-grid changes in treatment status over time (**?**). Each cohort includes only newly treated observations that start treatment in a particular month, together with units not yet treated and never treated. We build a separate database for each cohort of treatment and stack these datasets into a pooled sample. For a 5km^2 grid-cell i in assembly constituency j, month-year t, and cohort c, we estimate:

$$\text{Fires}_{\,ict} = \beta \text{Down} > \text{Up}_{ict} + \alpha_{ic} + \gamma_{jtc} + \delta X_{it} + \epsilon_{ict} \tag{1}$$

where our dependent variable is the number of fires in a grid month-year. The coefficient of interest, β, captures the effect of treatment, as described above, on the number of fires in that grid. The high-dimensionality of our data allows for the inclusion of fine-grained fixed effects. First, α_{ic} are grid-cell-cohort fixed effects that absorb time-invariant factors for over 76,800 grid-cells in our data, such as crop suitability or economic characteristics of small locations. Second, γ_{jtc} are jurisdiction by month-year-cohort fixed effects that account for time-varying shocks at the constituency level. X_{it} is a vector of grid-level controls that includes actual wind speed and direction. Given the stacked nature of our data, we interact each of the fixed effects with cohort fixed effects. Standard errors are clustered at the grid-cell and constituency by month-year level. The first accounts for temporal dependence of observations at the treatment status level (grid-cell). The second accounts for potential spatial correlation in wind patterns.

A typical concern in the standard difference-in-differences setting is that treatment might be assigned based on pre-existing characteristics of the unit endogenous to the outcome: for example, poorer areas might be more likely to receive welfare interventions. Our setting has instead a unique feature: treatment, which locations is more likely to be a source of pollution for a jurisdiction, is assigned based on changes in wind direction interacted with the shape of the assembly constituency polygon and its population, a quantity which is plausibly exogenous. This feature makes the parallel trends assumption —that treated grid-cells would have displayed a similar trend in fires as the control grid-cells had the wind not shifted, exposing a larger area of the jurisdiction to smoke— considerably more credible, as treatment is assigned quasi-randomly.

We nonetheless provide evidence consistent with the parallel trends assumption with a stacked-by-event event study. For relative time r from the switch to treatment, *Switch*, we estimate:

$$\text{Fires}_{\ ict} = \sum_{r=-n}^{N} \zeta_r Switch_{ir} + \alpha_{ic} + \gamma_{jtc} + \delta X_{it} + \epsilon_{ict} \tag{2}$$

using the same cohort structure and fixed effects as the main specification, where *Switch* is a dummy that takes value one when unit i is r periods away from its treatment switch, and zero otherwise. Our coefficients of interest ζ_r represents the effect of treatment status in each relative month before and after treatment, where the month prior to the switch is the omitted category. The event study differs from the table in focusing on a six-month window around treatment (Figure B.3). In all event studies, we adjust for the extrapolation of a linear trend from the pre-treatment periods, following common practice in recent difference-in-differences estimation (**?**, p. 2558). See for instance: **???**.

3.3 Results

Table 3a reports the main result. When wind is expected to blow smoke into a grid's assembly constituency rather than away from it, we observe 9.7 to 10.9 fewer fires per 1,000 in each grid-month, corresponding to a reduction of approximately 5.8 to 6.6% relative to the average number of fires when a grid is not a source of exposure. These results are consistent with two interpretations, both of which we classify under the broader umbrella of political accountability. First, it could reflect politicians enforcement effort toward the locations whose smoke would fall on their own voters, reflecting either constituents' direct demand or politicians' own calculus of citizen welfare. Another possibility is that farmers anticipate that enforcement will be concentrated where burning would pollute the constituency, and do not burn as a result. This is a deterrence channel represents an accountability incentive because the latter is what generates variation in behavior.

Figure 3b plots the event study coefficients. Before switching to polluting most of the constituency, grids that will switch experience a similar number of fires as others. From the month they switch to being a source of exposure for constituents, however, the number of fires significantly drops. This effect correspond to a 13% reduction in fires with respect to baseline. The result is mainly driven by rice-producing areas where we observe 33% less fires with respect to baseline (Figure B.7), confirming the main result that politicians are significantly more likely to reduce fires when pollution would affect their own constituency.

To give a sense of the public health stakes of this political action, we translate the 13 percent fire reduction into air quality and mortality estimates using atmospheric modeling from **?**. During the post-monsoon burning season, when rice-burning effects are concentrated, crop residue burning is estimated to contribute 26 μg m^{-3} to the mean population-weighted PM2.5 concentrations in India. A 13 percent reduction in fires corresponds to a reduction of approximately 3 μg m^{-3} in burning-season ambient PM2.5 at the national level, and 10-29 μg m^{-3} in the northwestern constituencies where burning is concentrated. In mortality terms, applying this proportional reduction to Lan et al.'s estimate that Punjab, Haryana, and Uttar Pradesh collectively account for approximately 70 percent of the 69,000 annual premature deaths attributable to crop burning in India would yield roughly 3,100 fewer premature deaths per year. These figures should be interpreted with the caveat that the estimated fire reduction may in part reflect spatial displacement across constituency lines

Change the numbers

rather than a net decrease in total emissions: if enforcement effort in treated areas is offset by reduced effort elsewhere in the constituency, aggregate pollution and mortality benefits would be correspondingly smaller. Nonetheless, this estimate provides a useful benchmark for the scale of impact that would be achievable if politicians were incentivized to act uniformly as they do in the places and times where their accountability incentives are heightened. Appendix B.3 details the derivation of these figures.

Robustness: We start by testing whether the findings are robust to adopting different definitions of treatment. In Table B.4, we define a grid-cell as treated when most of the area of the constituency (rather than most of its population) lies downwind from the grid, finding effects of similar magnitudes and significance (column 2); we standardize the independent variable, instead of dichotomizing it, so that the treatment coefficient can be interpreted as the effect of a one standard deviation increase in the size of the downwind area (column 3); we adopt two other continuous definitions of treatment, the share of assembly constituency area that is downwind (column 4) and the difference between downwind and upwind area (column 5). In Table B.5 we vary the definition of the dependent variable, replacing the count of fires with a dummy equal to one when any fire took place in the grid-cell (column 1); with the natural logarithm of the count of fires plus one (column 2); and with a continuous variable capturing the average brightness of fires in a grid-month, a measure of fire intensity (column 3). Results are robust in all cases. When we disaggregate the analysis by year, the effect remains negative in each of the 10 years in our sample, with more imprecisely estimated effects (Table B.6). When we subset by state, the magnitude of the effect varies in line with expectations: the drop in fires is largest in Haryana (14%), Punjab and Uttar Pradesh (5 and 4% respectively), where agriculture is most prevalent and crop burning salient, while there is a positive effect in Bihar (Table B.7). We test the importance of differential pre-trends in Figure B.5 following the approach by **?**. Panel (a) presents the standard event-study estimates, while Panel (b) presents the de-trended event-study estimates, as in Figure 3b, following **?**. We follow **?** in also conducting sensitivity analyses for violations of the pre-trends assumption using the smoothness restriction (**?**), showing that findings are robust to violations of parallel trends up to 0.75 times the size of the highest observed pre-trend deviations (Figure B.5c).

Strategic action and inaction near constituency borders: Our next question is whether the main effect, a reduction in crop-related fires when constituents are exposed, is driven by politicians becoming more active where pollution harms their own constituency, by politicians becoming less active where pollution is displaced onto neighboring constituencies, or by both. To answer this question, we examine how fire activity varies with strategic considerations about pollution exposure.

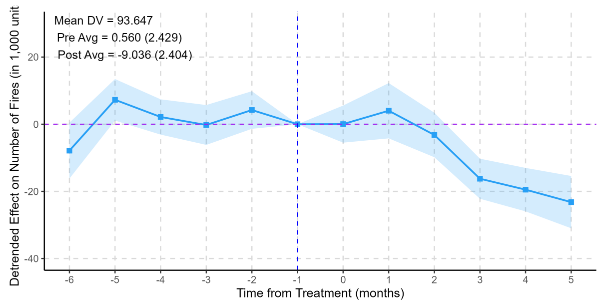
When burning occurs close to an upwind border, a larger share of the smoke remains within the politician's own constituency, strengthening incentives to intervene. When burning occurs close to a downwind border, the smoke is shifted into a neighboring constituency, weakening incentives to act. To test if politicians respond to both these incentives, we construct a new dataset at the level of grid-cell and each assembly constituency-border pair. Because wind direction varies over time, the same border can be downwind in some periods

Figure 3: **Effect of constituency exposure to smoke on number of fires**

(a) Main Table

	(1)	(2)	(3)	(4)
	Number of Fires (in 1,000 units)			
Down > Up Population	19.795***	-0.508	6.533***	-0.997
	(3.021)	(1.167)	(1.093)	(1.570)
Observations	63,550,570	63,550,441	62,551,714	62,462,012
N Assembly Constituencies	808	808	808	808
Month-Year FE				
AC FE				
AC × Month-Year FE				
Grid FE				
Mean DV	153.216	153.216	153.216	153.216

(b) Event-study relative to exposure



(c) Effect of constituency and district exposure to smoke on N Fires

	(1)	(2)	(3)	(4)	(5)
	Number of Fires (in 1,000 units)				
Down> Up Bureaucrat	-68.093***	-36.112***	-2.519***	-8.010***	-15.619***
	(6.180)	(1.023)	(0.761)	(0.679)	(0.773)
Down> Up Politician (Pop)	-2.720	-12.244***	5.999***	-6.781***	-3.613*
	(5.901)	(1.966)	(1.608)	(1.967)	(2.029)
Down> Up Pol. (Pop) × Down> Up Bur.	19.150***	12.824***	0.286	11.369***	1.236
	(6.732)	(2.294)	(1.770)	(1.879)	(2.106)
Observations	62,462,012	62,462,012	62,462,012	62,462,012	62,459,537
N Assembly Constituencies	808	808	808	808	808
N Districts	160	160	160	160	160
Month-Year FE	N	Y	N	N	N
AC FE	N	Y	N	N	N
AC × Month-Year FE	N	N	Y	Y	N
District × Month-Year FE	N	N	N	N	Y
Grid FE	N	N	N	Y	Y
Mean DV	151.037	151.037	151.037	151.037	151.037

Notes: (a) Results from Equation 1 of the effect of constituency exposure to smoke on the number of fires (in thousands) in a grid-cell and month. The treatment indicator *Down>Up* is a dummy equal to one in months and grids from which most of the assembly constituency's population is exposed to smoke (i.e. when the constituency's population in the downwind area is larger than the population in the upwind area from the grid). The table reports the mean of the dependent variable when treated grids are upwind. All regressions control for average wind speed and direction. Standard errors are clustered at the grid and assembly constituency × month-year level. *(b)* Results from Equation 2. The reference category is t=-1, i.e. the month before the switch. We use a stacked diff-in-diff to generate cohorts based on switching date. We restrict the analyses to a 6 months window around the switch to maximize support. The control grids are never treated grids and grids that are upwind around the switching date. The regression includes grid and assembly constituency × month-year × cohort fixed effects and controls for average wind speed and direction. Standard errors are clustered at the grid × cohort and assembly constituency × month-year × cohort level. *(c)* Results from Equation 1 with the additional interaction with a dummy (*Down > Up Bureaucrat*) equal to one in months and grids from which most of the district area is exposed to smoke, i.e. the area of the district downwind is larger than the area upwind from the grid. The table reports the mean of the dependent variable in grids that are upwind with respect to the borders of both the constituency and the district. All regressions control for average wind speed and direction. Standard errors are clustered at the grid and district × month-year level.

and upwind in others. We calculate the distance from each grid-cell to each relevant border and use this time-varying proximity to examine whether politicians adjust their behavior strategically as opportunities to internalize or externalize pollution become stronger.

The results in Figure 4 reveal two patterns. First, fires decline monotonically as one approaches the upwind border, consistent with politicians acting to reduce fires when a larger share of their own constituency would bear the pollution costs. Second, fires increase as one approaches the downwind border, consistent with reduced effort when pollution can be offloaded onto neighboring constituencies.

Together, these results show that politicians behave strategically along both margins: they increase effort when their own constituents are more exposed, and they relax effort when the costs can be shifted elsewhere. Additionally, politicians are also acting strategically in increasing the intensity of their action (and inaction) depending on the intensity of pollution exposure —distance to the border.

3.3.1 Alternative Interpretations

Farmers' Pro-sociality: A first alternative explanation is that fires may be reduced by farmers themselves, rather than by politicians, because farmers behave prosocially toward nearby residents who would be exposed to the resulting pollution. Under this account, farmers would care about neighbors as such, regardless of whether they belong to the same electoral constituency, since the constituency is an administrative unit with no obvious social or cultural meaning.

To assess this possibility, we construct a placebo measure of exposure based on farmers' local social environment rather than political boundaries. Specifically, around each grid we draw a placebo circle with a radius of 13 km, corresponding to the median size of an assembly constituency, and use it to approximate the set of nearby residents potentially relevant to farmers' prosocial concerns. Figure B.4 illustrates this strategy. In this analysis, a grid is coded as treated when the population living downwind within the placebo circle is larger than the population living upwind.

3c Table B.8 reports the estimated treatment effect for the full sample (column 1), for locations that are treated both in the placebo specification and from the politician's perspective (column 2), and for locations that are treated in the placebo-farmer specification but are control areas from the politician's perspective (column 3). We find no evidence consistent with farmer prosociality. The reduction in fires appears only in locations where politicians have incentives to act, while in places where politicians have weak or no incentives, the estimated effect is even positive. This pattern suggests that the reduction in fires is not driven by farmers concern for downwind residents, but by political incentives operating through constituency boundaries.

Results now changed

Bureaucratic action: A second alternative explanation is that the main results are driven by bureaucrats' incentives rather than by politicians'. Indeed, in previous work we show that bureaucrats respond to crop burning when their own districts receive the brunt of air pollution versus their neighbors (**?**). The objective here is not to make the case that bureaucrat incentives do not matter; rather, we want to establish that political incentives also matter and explore the interaction of the two. To do so, we examine whether the effect of

Figure 4: **Effect of grid distance to the upwind (in blue) and downwind (in red) border on number fires**

Missing figure

Notes: (Blue) Effect of being close to an upwind border with another constituency, where smoke would affect the politicians' assembly constituency. (Red) Differential effect (from the upwind border) of being close to a downwind border, where smoke would travel to the neighboring jurisdiction. All effects are plotted by quintiles of distance to the border, and the 5th quintile, the farthest, is used as omitted category. The regression includes grid and border $\times$ month-year fixed effects and control for average wind speed and direction. Standard errors are clustered at the grid and assembly constituency $\times$ month-year level.

pollution visibility depends on whether exposure is concentrated only within the politician's constituency or also within the jurisdiction of the relevant bureaucratic authority. In other words, we test whether politicians still reduce fires when the same pollution would affect not only to their own constituents but also district residents.

The results are reported in Table. We continue to find that both politicians and bureaucrats reduce fires when their respective jurisdictions are exposed to pollution from potential crop burning in a location but not vice versa suggesting that political and bureaucratic incentives matter on their own. However, we also find that the total effect of a location being a source of pollution for both the politician's constituency and the bureaucratic district is smaller, suggesting that the politician's incentive to reduce visible pollution is weakened once responsibility is shared with the bureaucracy. One interpretation is a coordination failure: when both political and administrative authorities are implicated, each may expect the other to bear the costs of enforcement, resulting in less action overall. The fact that both actors matter reflects the layered governance structure of Indian anti-burning policy, in which politicians and bureaucrats share overlapping responsibility.

4 Mechanism

4.1 Political Incentives Mobilize Government Machinery

The evidence so far shows how political incentives translate into control of polluting behavior, or lack thereof. In the context section, we discussed qualitatively how politicians can concretely affect crop burning by galvanizing the machinery of the state. We now turn to examining this quantitatively.

Outside of legislation, where political will directly lays the foundation for policy action, most policy change is not implemented directly by politicians and requires that they coordinate with the bureaucracy. Therefore, for outcomes like crime reduction or increases in public spending, direct evidence of politicians' role is difficult to isolate **??**.

While, the same is true of controlling crop burning, there are two sources of data that allow us a peek inside how political incentives may shape political action via the bureaucracy. First, from 2015-2016, politicians lead meetings with district administrators across India engage to coordinate on public service provision via the institution of the District Development Coordination and Monitoring Committee (DISHA) as we described above. These meetings and their outputs give us a rare window to observe evidence of political action. Second, India's eCourts data contains records of cases that relate to infringements of environmental laws, including crop burning. We can study how a change in political incentives impacts the production of these cases.

DISHA Data As the DISHA institution gives politicians monitoring authority over bureaucratic action, the content of its proceedings offers a direct window into what issues politicians choose to raise and pursue. We focus on whether politicians table the issues of agriculture and air pollution, a manifestation of the political will to act on crop burning through institutional channels.

These proceedings and their associated documents are posted across hundreds of separate district and state government websites, with no common catalog and no standard address. We therefore use a web-crawl strategy that scrapes state and district websites to retrieve any document or webpage mentioning DISHA, extract the location and date recorded inside each document, and code its topic, such as documents relevant to agriculture and air pollution. For example, one document we retrieve contains the proceedings of a DISHA meeting held in Kaimur, Bihar, chaired by Shri Chhedi Paswan, Member of Parliament for Sasaram, on 29 June 2018, which we assign to Kaimur district in June 2018. See Appendix A.3.2 for details of document retrieval and processing.

This procedure yields an unbalanced district-by-month panel of DISHA documents. To ensure that a zero reflects the absence of agricultural or air pollution content rather than the absence of any DISHA documentation, we include each district only from the first month-year in which at least one DISHA document is available.[21]

Courts Data We also study enforcement action against crop burning via India's eCourts records. Court cases are a measure of the state acting against crop burning, prompted by a combination of bureaucratic mobilization and the political pressure that activates it. Where DISHA captures what politicians choose to raise, court filings capture whether the enforcement apparatus is actually engaged.

We access judicial case records from the SHRUG database (**?**), which digitizes the Indian eCourts platform. We retain cases filed under the Air (Prevention and Control of Pollution) Act, 1981, the statute under which crop burning offenses are prosecuted. See Appendix A.3.1 for further details.

Anzony: please fill in exactly how we filter these cases (act code field, keyword search, or both), how many cases the filter returns, and whether we can

Our outcome is case filing that measures whether enforcement is initiated, signaling the transfer of a case from police to the judicial system, which is the margin politicians plausibly influence. Filings also avoid the long and variable lags between filing and judgment in Indian district courts. The sample runs from 2012 to 2018, the period covered by the SHRUG judicial extract.

Empirical Strategy We test whether DISHA meetings and court cases respond to the same political incentives that reduce fires. Both outcomes are recorded at the district level: DISHA meetings are convened by district, and eCourts records the location of the court rather than the location of the offense. We construct both as district-month panels and link them to our assembly constituency-month database by mapping each constituency to the district that contains it. 30 out of 853 assembly constituencies in our sample (3.5%) span more than one district. Because documents and judicial case records are identified only at the district level, matching them to these constituencies would be ambiguous, and we therefore drop them from the analysis.

As we are interested in understanding politician level decisions, we aggregate our grid-level treatment into a continuous, constituency-level measure of the incentive a politician faces, which we call *Potential Exposure*, capturing how much of a constituency's population could be affected by smoke from crop burning. As above, let i index 5km^2 grid-cells and j

[21] Prior to a district's first documented meeting, a zero would be ambiguous: it could reflect that the issue was not raised, or simply that DISHA meetings had not yet been initiated or documented in that area.

assembly constituencies, and let Down_{it} denote the constituency population lying downwind of grid i in month-year t, the same quantity that defines $\text{Down} > \text{Up}_{it}$. Let Pop_j the constituency's total population. We define:

$$\text{Potential Exposure}_{jt} \;=\; \frac{1}{\text{Pop}_j} \sum_{i \in j} \text{Down}_{it}, \tag{3}$$

The interpretation is that $\text{Potential Exposure}_{jt}$ is the average number of times each constituent inside the politician's own constituency *potentially* receives pollution from an upwind fire. We standardize $\text{Potential Exposure}_{jt}$ over the estimation sample (mean = 60.12, SD = 37.71), so a one-unit increase corresponds to a one-standard-deviation increase in the number of own-constituency potential pollution for an average constituent.

For outcome Y in constituency j and month-year t, we estimate

$$Y_{jt} \;=\; \lambda\,\text{Potential Exposure}_{jt} + \mu_j + \nu_t + \xi X_{jt} + o_{jt}, \tag{4}$$

where μ_j are constituency fixed effects, ν_t are month-year fixed effects, and X_{jt} is a vector of controls containing the constituency's average wind speed and direction. Standard errors are clustered at the constituency level.

Results Table 4 presents the results. First, in Column 1, we show that a one standard deviation increase in citizens' potential exposure to pollution decreases actual fires by 36 percent ($p < 0.1$).

In columns 2 and 3 we explore DISHA outcomes and show that a one-standard-deviation increase in exposure increases the probability of agriculture document by 70% relative to a baseline rate of 0.5 mentions per 100 constituency-months, and increases the probability of an environment document by 60% relative to a baseline of 0.3 mentions per 100 ($p < 0.1$ for both). These are large effects given how rarely these topics appear in DISHA documents absent accountability pressure.[22]

Turning to court cases in Columns 4 and 5, we find that a one-standard-deviation increase in politicians' incentives yields no significant effect within the same month. In the following month, however, exposure results in 0.053 additional court cases against crop burning per constituency-month, a 72% increase relative to the mean ($p < 0.01$). This delayed response is consistent with the time required for enforcement actions to materialize as cases are passed to the judicial system only after necessary evidentiary work by the police. Given that severe penalties are rarely issued (on average, less than one per district every month), this represents a substantively meaningful enforcement response.

Taken together, these results show evidence for the mobilization of government machinery in response to changes in political incentives. However, these estimates should be interpreted with two caveats. Because treatment varies continuously and over time, with units switching in and out of treatment, the design is too complex to permit formal validation of difference-in-differences assumptions. The low geographic granularity of the outcome data also means

[22] Appendix Table E.1 shows that this response is specific to agriculture and the environment because higher potential exposure does not increase discussion of other topics; if anything, documents related to welfare and development declines, suggesting that politicians reallocate their agenda toward agricultural and environmental concerns when constituent exposure to pollution increases.

that several coefficients are imprecisely estimated. Notwithstanding these limitations, the findings provide direct evidence that politicians translate accountability incentives into concrete institutional action, both by steering bureaucratic attention toward crop burning in coordination meetings, and by directing subsequent enforcement through the courts.

Table 1: Pollution Exposure, Politicians' Actions, and Electoral Consequences, Count × Downwind Pop

Missing table: tables/action˙final5

Notes: The table reports politicians' actions (columns 1-4) and electoral outcomes (columns 5-6) by level of constituency exposure. *Potential Exposure* is the sum across grids of the number of fires recorded in the same calendar month of the previous year, each weighted by the constituency population lying downwind of that grid, normalized by the constituency's total population, and standardized (z-scored) within the estimation sample. *Actual Exposure* is defined analogously using actual fires recorded in the constituency-month-year. In columns 6 and 7, both variables are averaged at the constituency-by-electoral-term level before standardization. Columns (1) and (2) use DISHA documents to report the probability that a document is recorded per 100 in a given month-year, for the following categories: agriculture (column 1) and environmental pollution (column 2). Columns (3) and (4) report the change in probability of filing court cases about pollution in the same month and in the following month, respectively. Column (5) reports the probability that the incumbent politician runs for re-election. Column (6) reports the incumbent's vote share in the following election. The table reports the mean of the dependent variable in the sample. All regressions in columns (1)-(4) control for wind direction and average wind speed. Standard errors are clustered at the constituency level.

4.2 Exposure to Fires Drives Electoral Outcomes

The previous subsection showed that politicians respond to accountability incentives by mobilizing state action against crop burning. To complete evidence on the way this accountability mechanism operates, we now show evidence on voters: constituents should reward politicians who reduce fires and punish those who do not. We examine this directly by asking whether constituents' *actual* exposure to pollution over an electoral term affects incumbent electoral performance.

Empirical Strategy To answer this question, we create a modified measure of exposure, which we call Actual Exposure$_j$. While for the first mechanism subsection we tracked month-by-month intensity in politicians' incentives to act using potential exposure, this new variable is constituency-by-electoral-term measure of the cumulative exposure of voters to actual fires.

Table 2: Pollution Exposure, Politicians' Actions, and Electoral Consequences, Any Fire × Downwind Population

Missing table: tables/action˙final6

Notes: The table reports politicians' actions (columns 1-4) and electoral outcomes (columns 5-6) by level of constituency exposure. *Potential Exposure* is the sum across grids of the constituency's population living downwind of a grid that experienced a fire in the same calendar month of the previous year, normalized by the constituency's total population, averaged at the constituency-by-electoral-term level, and standardized (z-scored) within the estimation sample. *Actual Exposure* is defined analogously using actual fires recorded in the constituency-month-year. In columns 6 and 7, both variables are averaged at the constituency-by-electoral-term level before standardization. Columns (1) and (2) use DISHA documents to report the probability that a document is recorded per 100 in a given month-year, for the following categories: agriculture (column 1) and environmental pollution (column 2). Columns (3) and (4) report the change in probability of filing court cases about pollution in the same month and in the following month, respectively. Column (5) reports the probability that the incumbent politician runs for re-election. Column (6) reports the incumbent's vote share in the following election. The table reports the mean of the dependent variable in the sample. All regressions in columns (1)-(4) control for wind direction and average wind speed. Standard errors are clustered at the constituency level.

Table 3: Pollution Exposure, Politicians' Actions, and Electoral Consequences, rural grid × Count × Downwind Pop

Missing table: tables/action˙final7

Notes: The table reports politicians' actions (columns 1-4) and electoral outcomes (columns 5-6) by level of constituency exposure. *Potential Exposure* is the sum across rural grids of the number of fires recorded in the same calendar month of the previous year, each weighted by the constituency population lying downwind of that grid, normalized by the constituency's total population, and standardized (z-scored) within the estimation sample. *Actual Exposure* is defined analogously using actual fires recorded in the constituency-month-year. In columns 6 and 7, both variables are averaged at the constituency-by-electoral-term level before standardization. Columns (1) and (2) use DISHA documents to report the probability that a document is recorded per 100 in a given month-year, for the following categories: agriculture (column 1) and environmental pollution (column 2). Columns (3) and (4) report the change in probability of filing court cases about pollution in the same month and in the following month, respectively. Column (5) reports the probability that the incumbent politician runs for re-election. Column (6) reports the incumbent's vote share in the following election. The table reports the mean of the dependent variable in the sample. All regressions in columns (1)-(4) control for wind direction and average wind speed. Standard errors are clustered at the constituency level.

Table 4: Pollution Exposure, Politicians' Actions, and Electoral Consequences, rural grid × Any Fire × Downwind Population

Missing table: tables/action˙final8

Notes: The table reports politicians' actions (columns 1-4) and electoral outcomes (columns 5-6) by level of constituency exposure. *Potential Exposure* is the sum across rural grids of the constituency's population living downwind of a grid that experienced a fire in the same calendar month of the previous year, normalized by the constituency's total population, averaged at the constituency-by-electoral-term level, and standardized (z-scored) within the estimation sample. *Actual Exposure* is defined analogously using actual fires recorded in the constituency-month-year. In columns 6 and 7, both variables are averaged at the constituency-by-electoral-term level before standardization. Columns (1) and (2) use DISHA documents to report the probability that a document is recorded per 100 in a given month-year, for the following categories: agriculture (column 1) and environmental pollution (column 2). Columns (3) and (4) report the change in probability of filing court cases about pollution in the same month and in the following month, respectively. Column (5) reports the probability that the incumbent politician runs for re-election. Column (6) reports the incumbent's vote share in the following election. The table reports the mean of the dependent variable in the sample. All regressions in columns (1)-(4) control for wind direction and average wind speed. Standard errors are clustered at the constituency level.

For each constituency-term in office o, let M_o denote the number of months in the term. We define:

$$\text{Actual Exposure}_{jo} = \frac{1}{M_o}\sum_{t\in o}\left(\frac{1}{\text{Pop}_j}\sum_{i\in j}\text{Down}_{it}\times \#\text{Fires}_{it}\right), \tag{5}$$

where M_o is the number of months in term o, Down_{it} equals the number of people living downwind of grid i in month-year t, and $\#\text{Fires}_{it}$ is the number of fires in grid i in that month, such that when there are no fires, the quantity inside the parenthesis equals zero.

Actual Exposure_{jo} therefore represents the average monthly fires a citizen is actually exposed to in the constituency. As before, we regress incumbent electoral outcomes on the standarized version of this actual exposure (mean = 1040.09, sd = 2973.72), using constituency and state times election year fixed effects, and clustering standard errors at the constituency level.

Results We find that constituents exposed to more fires during an electoral term punish their incumbent in Table 4, Columns 6 and 7. A one-standard-deviation increase in accountability-relevant fire exposure reduces the probability that the incumbent chooses to run again by 0.054 percentage points, a 57% decrease relative to the baseline rerunning rate of 0.094 ($p < 0.1$). The same increase in exposure reduces incumbents' unconditional vote share by 2.1 percentage points, a 61% decrease relative to the mean at baseline ($p < 0.1$), though this decrease is mostly driven by the reduced re-contestation.

These effects indicate that when voters are exposed to more pollution from fires during when their politician had clear accountability incentives, they respond by withdrawing electoral support validating the mechanism underlying the main analysis: politicians suppress fires because failure to do so is electorally costly.

5 Countervailing Incentives

The first part of the paper provided evidence that there are prospects for environmental accountability. In this second part, we consider how accountability can weaken or break down with two fundamental levers that can act as countervailing incentives: organized lobbying, and adverse political selection.

5.1 Lobbying

From June 2020 to December 2021, India experienced the largest protest movement in the world history: 25 million farmers mobilized in large-scale protests against agricultural laws that aimed to deregulate the sale of farm produce, allow contract farming, and remove stockholding limits. One key reason for mobilization was the request to decriminalize crop burning, removing strict penalties such as high fines and jail terms ((**?**)). Farmers staged demonstrations, road blockades, and sit-ins, particularly in Punjab, Haryana, and at the borders of New Delhi. The demonstrations were successful: they led to the repeal of the laws and to the depenalization of stubble burning. This section analyzes the impact of all 390 farmers' protests events on crop burning, and asks whether those counteract accountability incentives. While those are large scale protests, we use the term 'lobbying' to refer to the organized pursuit of policy concessions by groups with a concentrated economic interest in a policy, even when this interest conflicts with the diffuse welfare of broader populations, consistent with canonical examples from the literature (**?**), from taxi driver strikes to energy sector mobilization, in which sector-based collective action secures concessions at public expense.

5.1.1 Data

We gather information on the date and location of each protest event related to the Farmers' Movement from June 2020 to December 2021 from **?**. We identify farmers' protests by searching for any conjugation of the word "farm" in the event description, considering both type and participants to the event. We validate this process by reading the full description for a random sample of events. Table C.1 present the descriptive statistics for the sample. Figure C.1 shows the temporal distribution of protest occurrence, highlighting a peak in late 2020 after the passage of the farm laws (September 2020). Protests were highly prevalent in Punjab but widespearead across all ares in our sample (Figure 5a).

5.1.2 Empirical Strategy

Treatment Definition: We code proximity to a farmer protest by drawing a 5km-radius buffer areas around the location of the 390 protests occurring between June 2020 and December 2021. We code a grid-cell as treated if it falls within at least one buffer area, from the month-year when a protest first occurs until the end of our sample period.[23] Figure C.2 provides a visual illustration of the grid-cells coded as treated in a constituency.

Estimation Strategy: We estimate the impact of proximity to a protest event using a stacked-design following **?**. For each event-time cohort in our data, defined as the month-year a location first experience a protest, we build one cohort specific dataset. Each dataset includes only never treated units –grids that never experienced a protest– and units first treated in that event-time –grids experiencing their first protest, eliminating the problematic comparison in case of heterogeneous treatment effects. We then stack these cohort-specific datasets into a pooled database. For grid-cell i in constituency j in province p, relative time from treatment r and electoral cycle s in cohort dataset c, we estimate:

$$\text{Fires}_{icr} = \beta \text{Post} \times \text{Protest}_{icr} + \alpha_{ic} + \gamma_{rc} + \theta_{p \times s} + X_{ipr} + \epsilon_{icr} \tag{6}$$

where the dependent variable is the number of fires per 1,000 units. We include a comprehensive set of fixed effects which account for cohort-specific and constituency specific factors. First, α_{ic} are grid $\times$ cohort fixed effects that absorb time-invariant factors for each grid in our sample within each sub-experiment. Second, γ_{tc} are relative time from treatment defined for every cohort that account for time-varying shocks at the national level, including changes in the legislative environment following protest events. Third, θ_{ps} are province $\times$ cohort -specific electoral cycles fixed effects that account for differences across cycles within a province. X is a vector of controls that includes province $\times$ cohort linear time trends, average wind speed and direction. While the stacked dataset is constructed at the month-year level, we aggregate relative-time into years (12 months from treatment) when plotting the event-study estimates to reduce noise. Standard errors are clustered at the buffer area level.

5.1.3 Results

After the first protest occurs in a given area, we observe 0.055 more fires per grid-month, a 13% increase with respect to an average of 0.439 fires in treated areas before any protest occurs (Table C.2, columns 1-3). This result is in line with politicians relaxing environmental control in response to concentrated farmer pressure. Figure 5c plots the de-trended event-study results showing an increase in the number of fires after the first protest. This effect corresponds to 21% more fires with respect to baseline. This indicates that politicians relax environmental enforcement in response to farmers organized protests.

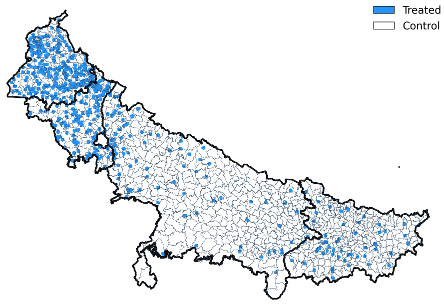
Figure C.3 presents the test of the importance of differential pre-trends, following **?**. Panel (a) presents the standard event-study estimates, while Panel (b) presents the de-trended estimates. Panel (c) presents the sensitivity analysis for violations of the parallel-trends assumption using the smoothness restriction of **?**, showing that our findings are robust to violations of parallel trends of more than 1 times the size of the largest observed pre-trend deviation.

Do we have what's the max of M? Do we need it if

[23] We consider as treated grid-cells that fall within a buffer area regardless of whether they fall inside or outside the constituency boundaries. Grids located in overlapping buffer areas are assigned to the buffer associated with the earliest protest.

Figure 5: **Effect of farmers' protests occurrence on number of fires**

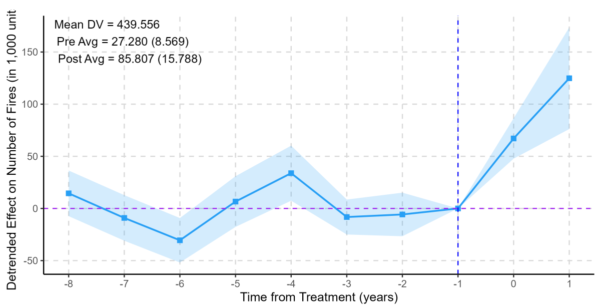
(a) Protest occurrence by location



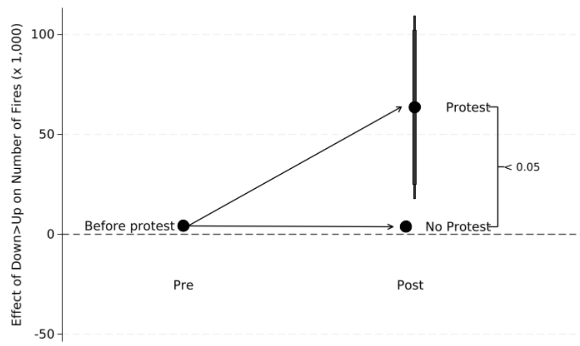
(b) Farmers' Protest

Missing figure

(c) By relative time period



(d) By constituency exposure to smoke



Notes: (a) The map shows treated (blue) and control (white) grid-cells. A grid-cell is coded as treated if it falls within 5km-radius buffer areas around protest locations. Treatment begins in the month-year of the first protest within each buffer area. Control grid-cells are located outside all buffer areas. *(b)* This photograph is called "Art, pen and people". It was taken in the 2020 Indian farmers' protest by opposing Modi's government farm laws (**?**). *(c)* Event-Study results from Equation 6 . The reference category is t=-1, a year before the protest occurrence. The regression includes grid-cohort, and province electoral cycle fixed effects, and controls for province linear time trends, average wind speed and direction. Standard errors are clustered at the buffer area × assembly constituency level. The control group includes never treated grids only. *(d)* Results from Equation 6 with the additional interaction of a dummy equal to one in months and grids from which most of the assembly constituency area is exposed to smoke ($Down > Up$, see Section 3.2 for details). Plotted coefficients report the effect of being in a constituency exposed to smoke in the pre- vs. post-protest period, separately for constituencies that did and did not experience protests. The reference category is non-protest areas, before protests when they are upwind ($Down > Up$=0). Table C.2 (columns 4-6) reports the corresponding regression coefficients. The regression includes grid-cohort, year-cohort and province electoral cycle fixed effects, and controls for province linear time trends, average wind speed and direction. Standard errors are clustered at the buffer area × assembly constituency level.

Erosion of Accountability Incentives: Organized pressure from farmer lobbies might contribute to influence the accountability incentives we document in the previous section: do politicians continue reducing fires when smoke would impact their constituents, even after farmers' protests pressure them to stop sanctioning them for crop burning? Figure 5d explores this question by interacting the two treatments, the effect of a grid becoming a source of exposure for its constituency, and protests. Before protests, areas whose pollution burden falls on their own constituents experience fewer fires relative to areas that would pollute neighboring jurisdictions, in line with the accountability mechanism.

In protest-affected areas, the accountability effect disappears entirely after protests begin: fires increase significantly, offsetting the reductions in crop burning otherwise associated with political accountability. In places without protests, the accountability effect weakens, but this does not translate into higher crop-related fires relative to baseline. This pattern is consistent with organized lobbying increasing fires directly where protests occur, while also weakening politicians' incentives for environmental enforcement beyond the protest locations.

5.2 Adverse Selection

In India, candidates with direct ties to agriculture are often elected to state legislatures, especially in rural constituencies where farming remains economically and politically central. Once in office, these politicians may use their authority to pursue policies that are more aligned with the interests of their sector than with those of other constituents. We therefore examine whether the selection of politicians with agricultural ties shapes the incidence of fires after elections, and whether it alters the accountability incentives documented in the first part of the paper.

5.2.1 Data

We build new data on the sectoral identity of elected politicians in each Assembly Constituency in our sample by scraping information about politicians' profession and properties from Myneta.info (**?**). The website provides background information on candidates contesting elections from 2008, including profession, and details on movable and immovable assets. To capture politicians' self-interest in agriculture, we classify whether each politician's self-reported occupation is agriculture-related, using a large language model (GPT)[24] (See Section A.2 for additional details.). Our sample comprises 2,565 politicians, of whom 1,102 (43%) have agricultural professions. Table D.1 presents descriptive statistics for the resulting sample.

5.2.2 Empirical Strategy

Treatment Definition: A comparison of constituencies that ever elect agriculturalist politicians versus not would conflate politician characteristics with place characteristics. Rural localities with strong agricultural ties may be more likely both to elect agriculturalists and to experience more agricultural fires. To isolate the effect of electing an agriculturalist

[24]Most politicians in our sample report holding agricultural assets, leaving little variation to exploit. We therefore rely on self-reported profession as our primary measure of self-interest in crop burning.

from underlying place characteristics, we exploit within-location variation of winning politicians. We compare crop related fires before and after the same constituency switches from electing a non-agriculturalist to electing an agriculturalist.

Estimation Strategy: We estimate the impact of a constituency switching to electing an agriculturalist politician using a stacked design (**?**). For each switch — defined as a constituency electing an agriculturalist politician after not having one in the prior election— we build a cohort-specific dataset spanning 10 years (2 elections) around the event. Never-treated units, i.e. constituencies that elect a non-agriculturalist politician throughout the 10 year window, are the control group. To avoid compositional changes that might bias our estimates, each dataset is a balanced panel, including only units that we observe for the entire 10 year period. We then stack these cohort-specific datasets into a pooled database. For a 5km^2 grid-cell i in constituency j in province p, relative time from treatment r, election s in cohort-dataset c, we estimate:

$$\text{Number of Fires }_{icr} = \beta \text{ Post}\times\text{Agriculturalist}_{icr} + \alpha_{ic} + \gamma_{tc} + \theta_{ps} + X_{ipr} + \epsilon_{icr} \tag{7}$$

where the dependent variable is the number of fires per 1,000 units; *Post* is a dummy that takes value one from the cohort-specific year onward; *Agriculturalist* is a dummy equal to one after the election of a politician with agricultural ties. We include the same set of fixed effects as in the protest specification, which account for cohort-specific and electoral factors. In particular, the grid x cohort fixed effects guarantees that the estimate is identified from within-location changes over time, rather than from comparing potentially different constituencies that never elect agriculturalist politicians. The province × cohort × electoral cycle FEs (θ_{ps}) absorb the state-level political environment in each legislature term. For example, if BJP swept Uttar Pradesh in 2017, nominated more agricultural candidates across the state, and also changed state-level burning policy, that is absorbed. While the stacked dataset is constructed at the month-year level, we aggregate relative-time into years (12 months from treatment) when plotting the event-study estimates to reduce noise. Standard errors are clustered at the assembly constituency × election year level.

5.2.3 Results

When a constituency switches to electing an agriculturalist politician, fires increase by 6% with respect to baseline, corresponding to about 7 additional fires per grid-year as shown in Table D.2, Column 3. The effect is driven by rice-producing areas, where we observe 50% increase following the election of a politician with agricultural ties (Figure D.2). The de-trended event study in Figure 6c confirm this pattern: fires are 23% higher after a constituency switches into the electoral term of a agriculturalist politician, relative to the control group. Politicians with agricultural ties behave in line with the interest of their professional group, avoiding enforcement pressure on environmental damage. Figure C.3a the standard event-study to test for the importance of differential pre-trends, following **?**. Figure C.3c shows the sensitivity analysis for violations of the parallel-trends assumption using the smoothness restriction of **?**, showing that our findings are robust to violations of parallel trends of 0.75 times the size of the largest observed pre-trend deviation.

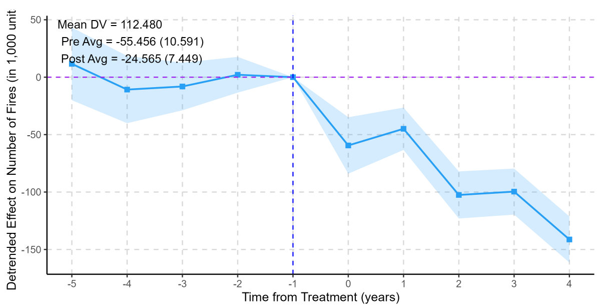
Figure 6: **Effect of switching to an agriculturalist politician on number of fires**

(a) Treatment status by grid over time

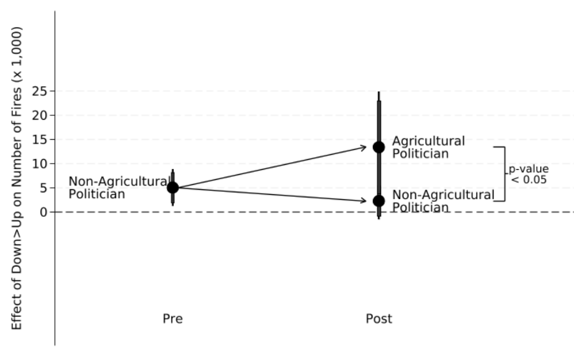
(b) My Neta: Profession

Missing figure

(c) By relative time period



(d) By constituency exposure to smoke



Notes: (a) The plot reports treatment status over time at the grid level. Each row represents a grid, sorted by first treatment time, while each column represents a month-year. Dark blue cells highlight the periods in which agrid had a self-reported agriculturalist politician in power. *(B)* Figure B shows an example of a Self-Declared Agriculturalist Politician report in My Neta Website. *(c)* Event-Study Results from Equation 7. The reference category is $t = -1$, one year before the election of an agriculturalist politician. The regression includes grid-cohort, and legislature fixed effects, and controls for province linear time trends, average wind speed and direction. Standard errors are clustered at the buffer area $\times$ assembly constituency level. The control group consists of never-treated grids over the period of analysis —that is, assembly constituencies, whether inside or outside the province holding the election, whose politicians' self-reported occupation is not agriculture-related in the electoral terms just before and after the election. *(d)* Results from Equation 7 with the additional interaction for constituency exposure to smoke. $Down > Up$ is a dummy equal to one in months and grids from which most of the assembly constituency area is exposed to smoke (see Section 3.2 for details). Plotted coefficients report the effect of being in a constituency exposed to smoke in the pre- vs. post-protest period, separately for constituencies that did and did not elect an agriculturalist. The reference category is constituencies with a non-agriculturalist in power, before elections when they are upwind ($Down > Up$=0). Table D.2 reports the corresponding regression coefficients. The regression includes grid-cohort, relative time-cohort and electoral cycle fixed effects, and controls for province linear time trends, average wind speed and direction. Standard errors are clustered at the buffer area $\times$ assembly constituency level.

Erosion of accountability: Figure 6d examines whether the election of an agriculturalist politician erodes the accountability incentives documented above by estimating the joint interaction of the two treatments. Before elections, fires fall in areas where politicians' own constituents bear the pollution burden, consistent with the baseline accountability pattern. After elections, however, the effect depends on who is elected. In constituencies that continue electing non-agriculturalists, fires remain lower when pollution affects the politician's own constituents. In constituencies that switch to an agriculturalist, instead, fires increase relative to the control group. These results suggest that politicians with personal ties to agriculture undermine the responsiveness to constituent exposure documented in the first part of the paper, consistent with the selection of politicians whose private incentives are misaligned with those of the majority of their constituents.

6 Conclusions

Environmental governance is a hard case for democratic accountability: pollution generates concentrated benefits and diffuse harms, and voters rarely prioritize the environment. Against this backdrop, we find that politicians do curb fires, when wind patterns shift smoke toward their own constituents. This is, to our knowledge, among the first large-scale evidence that accountability incentives can generate environmental protection even in a setting with millions of dispersed polluters and limited monitoring.

Accountability, however, does not hold to the competing pressure of concentrated interests and adverse selection. We show that organized lobbying by the farmers' movement increased burning, overriding politicians' responsiveness to their own constituents' welfare. In addition, we document that the election of politicians with private agricultural interests increases burning, even where accountability incentives would otherwise reduce it. Taken together, these results show that democratic accountability can produce environmental regulation, but concentrated interests that benefit from pollution can undermine it.

These findings speak to institutional design. Whether anti-burning policy works in India depends not only on national mandates and enforcement capacity but also on political incentives. Interventions that sharpen the link between pollution exposure and political responsibility, such as real-time air quality monitoring and transparent attribution of pollution to specific jurisdictions, could strengthen the accountability channel we document (**?**). But the erosion we find from organized interests and adverse selection suggests that electoral mechanisms alone might not be enough to confront the large challenge of air pollution.

Finally, our results inform a broader question about whether democracies can manage environmental externalities. The evidence from South Asia's air pollution crisis is that the capacity for accountability exists: politicians respond when their constituents pay the costs of pollution. The challenge is that this response is easily overwhelmed by the political economy forces arrayed against it. Sustaining environmental protection in democracies requires not just accountability but institutions that can withstand the pressures that erode it.

Appendix

A Data Construction

A.1 Combining all the data at a 5 square km grid-cell level

We combine all the data into a single dataset at the $5x5km$ grid-cell level. Wind and fire grids do not overlap precisely: to reduce errors in the assignment of fires to wind grids, we split the $25km$x$25km$ wind grid-cells into squares of $5x5k$ each. This guarantees that fires are assigned to the wind grid-cell which contains most of the fire grid-cell. Splitting wind grids in smaller parts also allows us to address another concern: some grids might overlap across jurisdiction boundaries. By reducing their size, we minimize the amount of misattribution and we are able to assign each grid to the appropriate jurisdiction. Any remaining error should bias our findings toward zero, as errors are uncorrelated with the dependent and independent variables. In case of overlap, we assign a grid to the jurisdiction that contains a larger area of the grid. For crop data, we assign the pre-period crop production to each $5km^2$ grid-cell to classify what crops are produced in each location.

A.2 Coding Farmer Politicians

We scraped Myneta.info, an online data repository of election-related information, to collect data on politicians. The website contains information on politicians' profession and their asset declarations.

To identify agricultural ties, we employed a large language model (GPT) and fix the temperature to 0 for inference, as this reduces variance in the generation and helps reduce hallucinations. We supplied each profession description to the model and recorded whether it described any kind of agriculture-related job; if so, the politician is classified as having an agricultural profession. Formally, a politician is coded as being an agriculturalist if the profession description refers to a profession related to agricultural. We are able to construct the indicator for every winning candidate in our sample. Only for 50 out of 22,652 politicians check
the website does not provide any information.

For validation we carried out two exercises: First, we ran the classification procedure ten times for each observation and assigned the final category using the modal classification across the ten runs. Second, we manually reviewed a stratified random sample of 10 percent of observations (stratifying on LLM positives and negatives of farmer politicians), and found zero errors.

The following fixed prompt was supplied to GPT for the coding of politicians' agricultural assets and professions:

```
You are a research assistant interested in political capture in
   India. You are researching whether politicians with agricultural
   assets are more prone to taking actions that favor agricultural
   businesses. Your current task is to assess whether an individual
   owns agricultural assets.\ You will be provided a table with two
   columns. The table summarizes the immovable assets owned by the
   individual. The first column describes the type of immovable
   assets listed in the corresponding row. The Second column lists
   the immovable assets of the corresponding type. The individual is
    considered to own agricultural assets ONLY IF any are listed in
   the row corresponding to agricultural assets. If the value says
   \"Nil\", we can infer that the individual does not own any such
   assets.\ Provide a JSON blob with your assessment, structured as
   follows. First, the \"explanation\" field should include your
   reasoning for your assessment on whether the individual owns
   agricultural assets. Second, the "owns_agricultural_assets" must
   include a boolean, true if the individual is assessed to own any
   agricultural assets, and false otherwise.\ You must only use the
   information included in the row of agricultural assets; any other
    kind of asset should be ignored.
```

We provide additional information on the use of AI in section A.4.

A.3 Politicians' Actions

A.3.1 Court Cases

We gathered data from the SHRUG (**?**) database, which contains all judicial case records from the Indian eCourts platform. From this database, we extracted all judicial cases in which there was a violation of the Air Pollution and Control Act of 1981 which include crop burning offenses from 2012 to 2018. Because these cases only report the district and the month in which they were filed, we match them to our sample at the district and month-year level from September 2012 to December 2018, corresponding to the start of our main sample period and the end of the available court data, respectively.

A.3.2 DISHA Document Collection

Our objective is to retrieve, for each district in our sample, any document or webpage that records DISHA meetings, and to match them to our data based on the location and time information inside the documents. Because DISHA proceedings are posted on hundreds of separate district and state government websites, with no common catalogue and no standard address, collecting them at scale requires a purpose-built web-crawl rather than a manual search. The crawl proceeds in two stages: a fast discovery stage that locates the pages where DISHA documents are likely to live, and a content-aware stage that follows those pages to the documents themselves.

Candidate websites. We exploit the fact that Indian district portals, though not standardized, follow a small number of recurring conventions. For each of the four states in our sample, Punjab, Haryana, Uttar Pradesh, and Bihar, and each district within them, we generate a list of candidate addresses by combining the district name with sixteen URL templates observed across the `.nic.in` and `.gov.in` domains; examples include a dedicated `/disha/` path, a `/document-category/proceeding/` path, and a notices page queried for the term "disha". We add a spelling variant that removes hyphens from multi-word district names, so that a district such as Charkhi-Dadri is reached whether or not the website encodes the hyphen. This yields on the order of twenty candidate entry points for each district. We then verify and retain only pages that are real.

Content-aware, multilingual spidering. We then read these pages and follow their internal structure to the documents. We parse each page's markup and extract two kinds of links: direct links to files, identified by their PDF, DOC, or DOCX extension and including documents embedded through inline frames; and navigation links whose visible text or address contains a DISHA keyword. The keyword set is deliberately multilingual and includes the Hindi spelling of DISHA alongside English cues such as "Proceeding", "Minutes", "Agenda", and "District Development Coordination". Because many websites do not list the documents on the landing page but behind a "DISHA" or "Proceedings" tab, the crawler follows these keyword links one level deeper, restricting itself to the same website, and repeats the extraction there. This one-step spidering is what allows it to reach documents that a fixed-address scrape would miss, while the depth and same-domain limits keep the crawler bounded and prevents it from wandering across the wider web. Each discovered document is downloaded.

Coverage. We attempted 152 districts across the four states, and retrieved public DISHA pages for 142 of them; the remaining ten districts had no publicly accessible page as of the collection date. The crawl captured 3,072 documents and successfully downloaded 3,059 of them, with 13 restricted or unavailable, for a total of roughly 6.3 gigabytes. , distributed as 1,251 documents in Uttar Pradesh, 1,094 in Bihar, 393 in Haryana, and 321 in Punjab.

Matching and coding. We match each downloaded document to a constituency and a month-year based on the district and time information inside the document and the webpage that hosts it, and for each document we also code its topic. Because the crawl is deliberately inclusive, capturing every candidate document rather than risking a missed proceeding, this matching and coding step is where documents that are not genuine DISHA proceedings are filtered out.

@mamoor we need to know how they are filtered out.

Each district is included in the analysis sample starting from the month-year of its own first available document.

A.4 AI disclosure

We use large language models in two components of the data-construction process. No generative AI tools were used to formulate the research question, select the empirical strategy, estimate the statistical models, or interpret the results.

First, we used GPT API to classify politicians' self-reported profession descriptions obtained from Myneta.info. Each description was evaluated using a fixed prompt asking whether it referred to an agriculture-related profession that we then manually validate for a small sample. More informaiton about this process is present in Appendix Section A.2.

Second, we used Anthropic's Claude Code agents to construct a dataset from publicly available District Development Coordination and Monitoring Committee (DISHA) documents that were processed separately using OCR, including general-text and Punjabi/Gurmukhi OCR when necessary. Claude reviewed the OCR-extracted text in batches and applied predefined and fixed coding rubrics to: (1) identify each document's primary topic; (2) assign relevance scores for agriculture, the 2020–2021 farmers' protests and agricultural reforms, and air pollution or crop-residue burning; (3) infer the document or meeting date, prioritizing the meeting date over drafting or publication dates; and (4) classify the document genre, including meeting summaries, reports, orders, policy plans, instructions, and budget or planning tables. More information about data generation and identification of principal variables is present in Appendix Section A.3.2

B Politicians' Incentives to Control Fires

This section presents additional results for the main analysis. First, we present descriptives to the main variables (Table B.1), including the seasonality of fires and wind direction (Figure B.1). Second, we present the distribution of observations by relative time period from treatment (Figure B.3), and an illustration of the empirical strategy underlying the pro-sociality test (Figure B.4). We then turn to results, showing that our main estimates, presented in Figure 3, are robust to alternative definitions of treatment and outcome, to alternative samples, and to alternative explanations. We first vary the definition of treatment (Table B.4) and the dependent variable (Table B.5), then we subset by year (Table B.6) and province (Table B.7). We next report whether the effect is driven by farmers' prosociality, rather than politicians (Table B.8). We then replicate our main result using downwind and upwind area, rather than population, to define treatment (Figure B.6). Next, we examine heterogeneity by interacting treatment with constituency-level agricultural characteristics (Figure B.7). Finally, we report the event-study specification as in equation 2 (Figure B.5a), after de-trending for the pre- treatment period linear trend (Figure B.5b), along with the Honest DiD sensitivity analysis by **?** (Figure B.5c).

B.1 Empirical Strategy: Event-Study

We follow recent applied work in difference-in-differences estimation (**?**) adopting a neutral transformation to de-trend the event studies (**??**). This approach accounts for the possibility that treated and control units' outcomes may have evolved along different paths in the pre-treatment period. In particular, we follow **?** and fit a linear time trend using our estimated coefficients from the pre-treatment period, anchored at zero in the omitted category, and then extrapolate this trend into the post-treatment period. This approach accounts for potential violations of the parallel trends assumption in settings, such as ours, where the differential trend is linear.

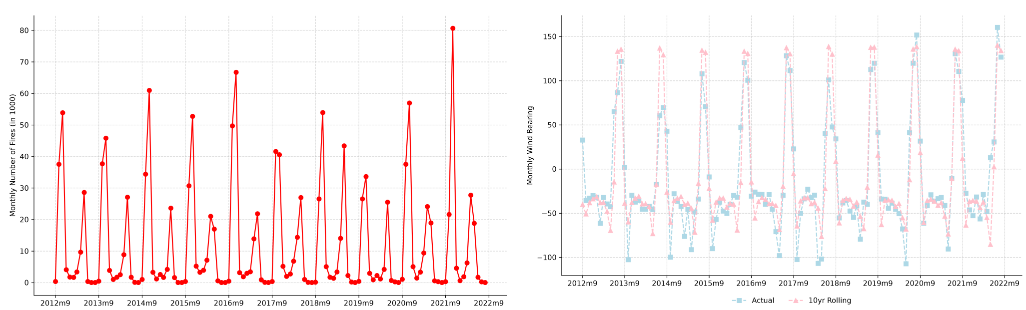
B.2 Descriptives

Table B.1: Descriptive Statistics

	Mean	SD	Min	Max	Observations	Unique Obs.
Grid ID						76,800
Year						11
Month						12
Assembly						808
Province						4
Number of Fires	0.155				9,216,000	
Down × Up AC Pop	0.508				9,216,000	
Average Wind Speed	1.317				9,216,000	
Wind Direction	-11.247				9,216,000	
Rice Production	0.457				9,216,000	

Notes: This table shows descriptive statistics of the main variable for the politician's incentive analysis

Figure B.1: Patterns of Seasonality in Fires Data



Notes: (Left) Total number of fires per month. (Right) District-average actual and 10 year rolling average wind direction. All data refer the states in our sample only —Bihar, Haryana, Punjab, and Uttar Pradesh— in the period September 2012 - August 2022.

Figure B.2: Rice Production in Northern Indian Subcontinent

Notes: Distribution of rice production (in tons) in 2010 across grids, classified by quintile of production.

Figure B.3: Number of observations per relative time period from treatment

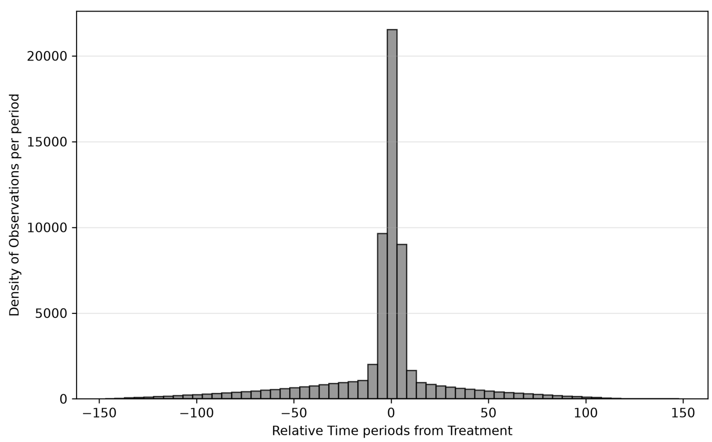
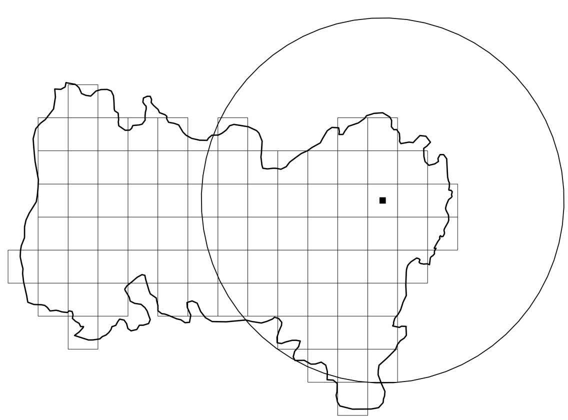


Figure B.4: Placebo border (circle) against the true constituency border for one grid



B.3 Translating Fire Reductions into Air Quality and Mortality Estimates

To translate our estimated fire reductions into air quality and mortality impacts, we draw on **?**, who use the GEOS-Chem adjoint atmospheric chemistry and transport model to quantify the sensitivity of India-wide population-weighted PM2.5 exposure to crop residue burning emissions from 2003 to 2019. Their approach computes how PM2.5 exposure responds to marginal changes in fire emissions at each location and time, making it well-suited to deriving the air quality consequences of the fire reductions we estimate. The model is validated against Central Control Pollution Board ground observations and NASA MODIS satellite-derived aerosol optical depth.

This translation rests on assuming proportionality between fire counts and emissions: a given percentage reduction in fire counts is treated as equivalent to the same percentage reduction in burning-attributable PM2.5 emissions and, in turn, in PM2.5 concentrations. We also assume that politicians the greater effort exerted by politicians in treated areas is a net effect. However, this could be compensated through reduced effort elsewhere in the constituency. As we note in the main text, this should thus be understood as a benchmark for what would be achievable under full increased political accountability, rather than as estimates of net gains. We give estimates for Punjab, Haryana, and Uttar Pradesh, the three states which Lan et al. identify as collectively responsible for over two-thirds of India-wide air quality impacts from crop burning. Numbers would be slightly larger adding the fourth state in our analysis sample, Bihar.

For the PM2.5 estimate, Lan et al. report that crop burning across India contributes an annual mean population-weighted PM2.5 of 6.7 μg m^{-3}, of which 64 percent arises from post-monsoon rice burning in October and November. This seasonal contribution of 4.3 μg m^{-3} to the annual mean, concentrated in approximately 60 days of burning activity, corresponds to an average burning-season PM2.5 from crop residue of approximately 26 μg m^{-3} across the India-wide population ($4.3 \times 365/60$). Applying our estimated 13 percent fire reduction yields a burning-season PM2.5 reduction of approximately 3.4 μg m^{-3} at the India national level. Residents of Punjab, Haryana, and Uttar Pradesh experience substantially higher burning-season concentrations due to their proximity to the source. Lan et al. document that burning-attributable PM2.5 in northern India during the burning season reaches 15 to 45 times the WHO annual mean guideline of 5 μg m^{-3}, implying local burning-season concentrations in the range of 75–225 μg m^{-3}, and that Delhi alone consistently exceeded 120 μg m^{-3} of PM2.5 averaged over the post-monsoon period across their full sample. A 13 percent reduction applied to these local levels implies reductions of roughly 10–29 μg m^{-3}.

For the mortality estimate, Lan et al. report 69,000 premature deaths per year attributable to ambient PM2.5 exposure from crop burning across India (95% CI: 57,000–80,000), with Punjab, Haryana, and Uttar Pradesh accounting for approximately 70 percent of this burden, or roughly 48,000 deaths annually. Applying a 13 percent proportional reduction to 48,000 yields approximately 6,200 fewer premature deaths per year. This figure is best understood as a full-accountability benchmark: it represents the hypothetical mortality reduction if political accountability suppressed fires by 13 percent uniformly across all grids and months throughout these states. In practice, our treatment variable (smoke directed toward politicians' own constituents) applies to approximately half of all grid-month observations, given a mean treatment rate of 50 percent in our sample (Table B.1). Weighting by the realized treatment share yields an estimate of approximately 3,100 averted deaths per year from the accountability mechanism as currently operating. The full-accountability figure is a benchmark for the potential public health impact of political accountability when fully operative, and when the increase in effort by politicians is not compensated by a decrease in effort elsewhere. Another caveat is that Lan et al.'s mortality estimates are averaged over 2003–2019; more recent years, which see higher burning intensity, would imply a larger baseline and correspondingly larger potential reductions.

B.4 Robustness

Table B.2: Effect of constituency exposure to smoke on number of fires, robustness by treatment indicator

downup_ac_pop	2.890**				
	(1.334)				
downup_ac		4.803***			
		(1.803)			
downup_1sd_pop			2.590		
			(1.619)		
down_percent_pop				-0.070*	
				(0.036)	
downup_diff_percent_pop					-0.035*
					(0.018)
Observations	63,530,478	63,530,478	63,530,478	63,530,478	63,530,478
N Assembly Constituencies	808	808	808	808	808
Mean DV	153.264	153.264	153.264	153.264	153.264

Notes: Results from Equation 1 considering different treatment indicators: a dummy for whether the downwind population is larger than the upwind population (as in the main specification) in Column 1; a dummy for whether the downwind area, rather than the population, is larger than the upwind area in Column 2; the a dummy for whether the downwind area is larger than the upwind area by one standard deviation or more in column 3; the share of downwind population over total population in Column 4; the percentage difference between down and upwind area in Column 5. The dependent variable is the number of fires in a 5 squared km grid-cell and month. The table reports the mean of the dependent variable when grids are upwind. All regressions control for average wind speed and direction. Standard errors are clustered at the grid and assembly constituency $\times$ month-year level.

Table B.3: Effect of constituency exposure to smoke on number of fires, robustness by treatment indicator

down_percent	-0.032***
	(0.007)
Observations	62,442,572
N Assembly Constituencies	808
Mean DV	153.264

Table B.4: Effect of constituency exposure to smoke on number of fires, robustness by treatment indicator

Down > Up (Pop)	-2.334
	(2.405)
Above Median Rice Production	0.000
	(.)
Down > Up (Pop) × Above Median Rice Production	2.768
	(5.416)
Observations	62,442,572
N Assembly Constituencies	808
Mean DV	153.264

Table B.5: Effect of constituency exposure to smoke on number of fires, robustness by dependent variable

	(1) Any Fire	(2) Log (N) Fires	(3) Mean Brightness
Down > Up (Pop)	-0.0008**	-0.0007	-0.2530**
	(0.0003)	(0.0004)	(0.1063)
Observations	62,462,012	62,462,012	62,462,012
N Assembly Constituencies	808	808	808
Grid FE	Y	Y	Y
Assembly × Month-Year FE	Y	Y	Y
Mean DV	0.05	0.059	16.548

Notes: Results from Equation 1 using alternative dependent variables: a dummy for whether any fire took place (Column 1), the natural logarithm of the number of fires plus 1 (Column 2) and the average brightness of fires in a grid-month – brightness is equal to zero when there are no fires (Column 3). The table reports the mean of the dependent variable when ever treated grids are upwind. All regressions control for average wind speed and direction. Standard errors are clustered at the grid and assembly constituency × month-year level.

Table B.6: Effect of constituency exposure to smoke on number of fires, robustness by year

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Number of Fires (in 1,000 units)									
	2012/2013	2013/2014	2014/2015	2015/2016	2016/2017	2017/2018	2018/2019	2019/2020	2020/2021	2021/2022
Down>Up (Pop)	5.381	0.673	9.912*	-2.800	1.007	-3.136	-12.267*	-1.449	-4.900	1.682
	(5.205)	(5.654)	(5.147)	(5.466)	(5.742)	(5.773)	(6.264)	(4.138)	(6.452)	(6.304)
Observations	5,552,535	5,876,789	6,007,388	6,194,026	6,280,088	6,250,728	6,479,638	6,352,381	6,417,605	6,201,762
N Assembly Constituencies	808	808	808	808	808	808	808	808	808	808
Grid FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Assembly × Month-Year FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Mean DV	151.139	141.492	145.732	151.219	180.108	151.672	162.715	104.452	168.627	175.076

Notes: Results from Equation 1 subsetting by year, from each September to the following August. The table reports the mean of the dependent variable in each year when ever treated grids are upwind. All regressions control for average wind speed and direction. Standard errors are clustered at the grid and assembly constituency × month-year level.

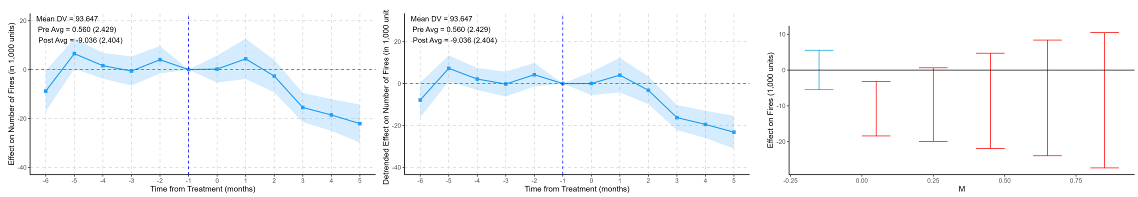
Table B.7: Effect of constituency exposure to smoke on number of fires, robustness by province

	(1)	(2)	(3)	(4)
	Number of Fires (in 1,000 units)			
	Bihar	Haryana	Punjab	Uttar Pradesh
Down>Up (Pop)	-0.385	10.526*	-0.340	-6.051***
	(2.457)	(6.034)	(7.163)	(1.336)
Observations	6,722,326	5,956,374	9,234,568	40,548,744
N Assembly Constituencies	237	84	107	380
Grid FE	Y	Y	Y	Y
Assembly × Month-Year FE	Y	Y	Y	Y
Mean DV	50.911	155.027	764.769	47.208

Notes: Results from Equation 1 subsetting by province. The table report the mean of the dependent variable in each province when ever treated grids are upwind. All regressions control for average wind speed and direction. Standard errors are clustered at the grid and assembly constituency × month-year level.

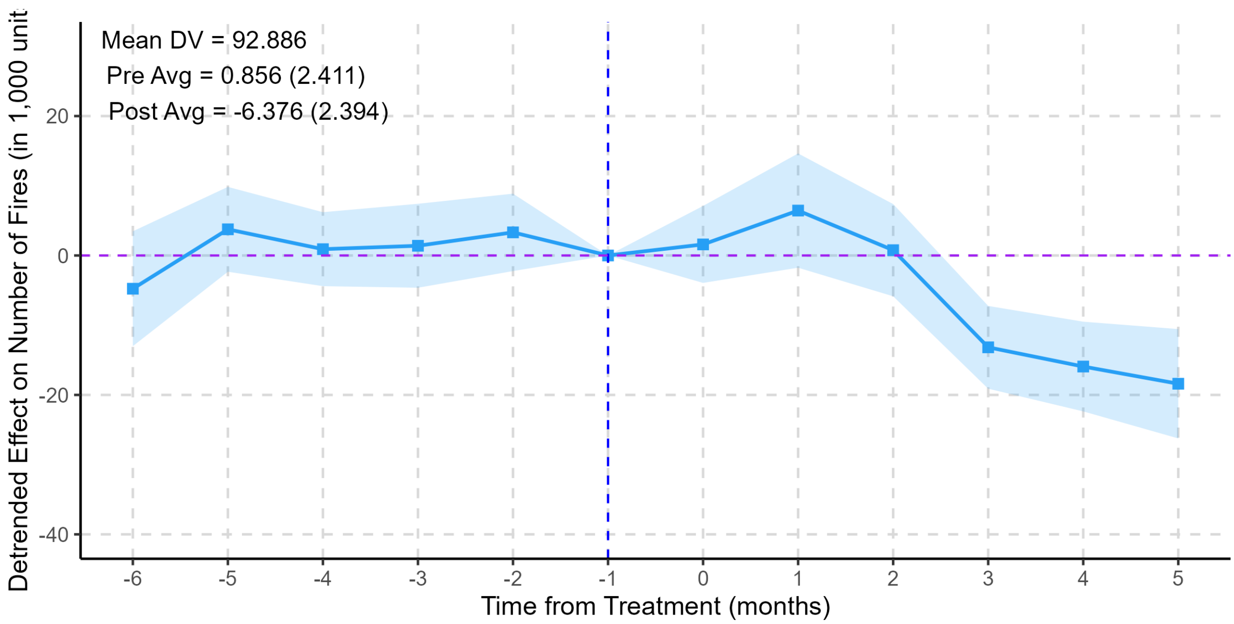
Figure B.5: Effect of constituency exposure on number of fires

(a) Event-Study (b) De-Trended Event-study (c) Honest DiD



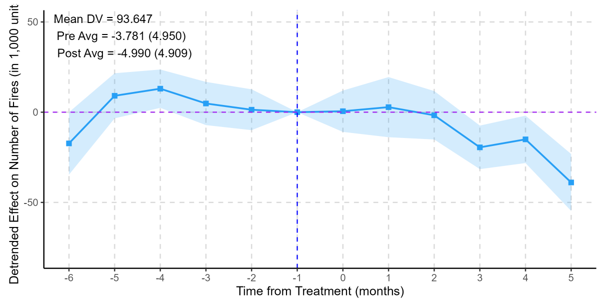
Notes: (a) Results from Equation 2. (b) Results from Equation 2 after de-trending for the pre-treatment period linear trend following **?**. (c) Sensitivity analysis for the average post-treatment effect allowing for violations of parallel trends (**?**). The reference category is $t = -1$,i.e. the month before the switch. All regressions include grid × cohort and assembly constituency × month-year × cohort fixed effects and control for average wind speed and direction. Standard errors are clustered at the grid × cohort and assembly constituency × month-year × cohort level.

Figure B.6: Effect of constituency exposure to smoke on number of fires, by area



Notes: Results from Equation 2. The treatment indicator *Down>Up* is a dummy equal to one in months and grids from which the majority of the assembly constituency's area is exposed to smoke with respect to the grid. The reference category is t=-1, i.e. the month before of the switch. We use a stacked diff-in-diff to generate cohorts based on switching date. We restrict the analyses to a 6-months window around the switch to maximize support. The control grids are never treated and upwind grids around the switching date. The regression includes grid and assembly constituency $\times$ month-year $\times$ cohort fixed effects and controls for average wind speed and direction. Standard errors are clustered at the grid and district $\times$ month-year level.

Figure B.7: Effect of constituency exposure to smoke on number of Fires, by constituency with rice production



Notes: Results as in Figure 3b with additional interactions by constituency agricultural characteristics. The figure plots the interaction term between treatment and an indicator for constituencies with above-median rice production. The reference category is t=-1, i.e. the month before of the switch. All regressions control for average wind speed and direction. Standard errors are clustered at the grid and assembly constituency × month-year × corhot level.

Table B.8: Effect of exposing the population in a farmer placebo area to smoke on N fires

Down>Up (13 km placebo)	4.899***	-3.909	5.863**
	(1.546)	(2.547)	(2.592)
Observations	133,548,526	64,854,771	67,925,304
N Assembly Constituencies	808	808	808
Grid FE	Y	Y	Y
AC × Month-Year × Cohort FE	Y	Y	Y
Mean DV	182.464	182.464	182.464

Notes: Results from Equation 1 considering a different treatment indicator: a dummy equal to one in months and grids from which the population downwind is larger than the population upwind in a 13km radius circle around each grid (median size of an assembly constituency). Column 1 considers the entire sample; Column 2 restricts to areas that are treated both for the farmer and the politician, i.e. downwind both with respect to the placebo area and the constituency; Column 3 restricts to areas that are treated for the farmer only, i.e. downwind with respect to the placebo area but upwind to the constituency. The table reports the mean of the dependent variable when treated grids are upwind. All regressions control for average wind speed and direction. Standard errors are clustered at the grid and assembly constituency × month-year level.

C Lobbying

This section presents additional results for the lobbying analysis. First, we present descriptive statistics for the main variables (Table C.1) and illustrate the construction of the empirical strategy (Figures C.1 and C.2). Second, we present the de-trended event-study along with the Honest DiD sensitivity analysis following **?** (Figure C.3). Lastly, we report heterogeneity by constituency exposure to smoke (Table C.2).

C.1 Descriptives

Table C.1: Descriptive Statistics

	Mean	SD	Min	Max	Observations	Unique Obs.
Grid ID					163,523,814	76,800
Year					163,523,814	11
Month					163,523,814	12
Assembly Constituency (AC)					163,523,814	808
Province					163,523,814	4
Within 5 km of Protest					163,523,814	2
Cohort					163,523,814	19
Legislature					163,523,814	12
Relative year	-4.013				163,523,814	13
Protest	0.001				163,523,814	2
Number of Fires (in 1,000 units)	134.395				163,523,814	45
High Rice Production (AC level)	0.437				163,523,814	2

Notes: This table shows descriptive statistics of the main variable for the protest analysis

C.2 Empirical Strategy

Figure C.1: Number of protests over time

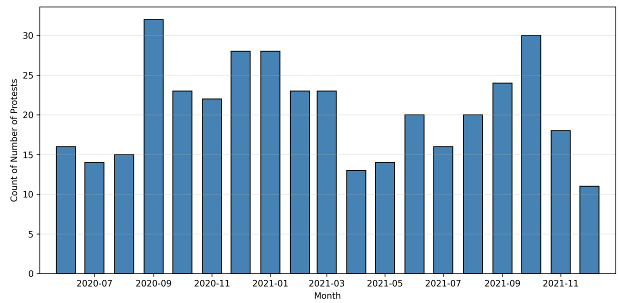
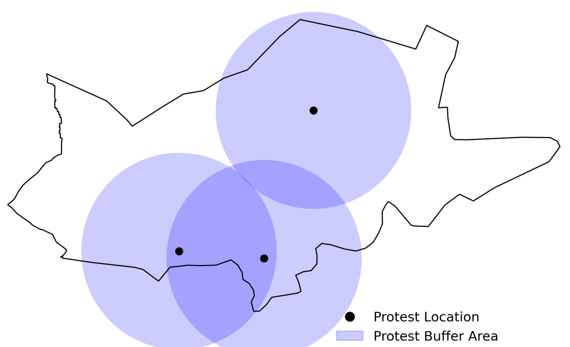


Figure C.2: Buffer area against Assembly Constituency border



Notes: The map plots the borders of Jind Assembly constituency (black line). Three protests occurred in this constituency during the sample period. Around each protest location, we construct 5km-radius buffer areas (blue shaded) around the protest location (black dot). We classify grid-cells falling within these buffer areas as treated, regardless of whether they fall inside or outside the assembly constituency boundaries. Grid-cells located in overlapping buffer areas (darker shaded) are assigned to the buffer associated with the earliest protest.

C.3 Robustness

C.4 Additional Results

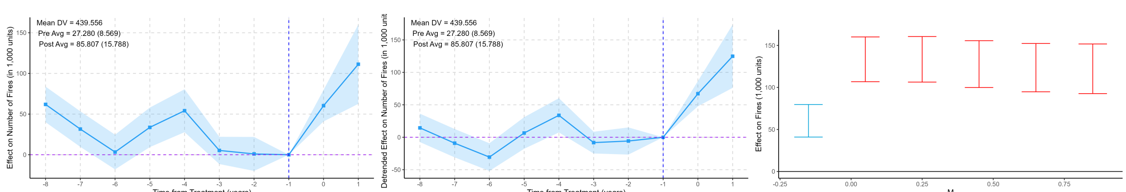
Table C.2: Effect of Protest occurrence on number of fires, by constituency exposure to smoke

	(1)	(2)	(3)	(4)	(5)	(6)
	Number of Fires (in 1,000 units)					
Post × Protest	15.614	82.424***	74.022***	22.620	89.245***	80.728***
	(11.924)	(10.920)	(10.813)	(15.853)	(14.963)	(14.897)
Down > Up (Pop)						
Post × Down > Up (Pop)						
Protest × Down > Up (Pop)						
Post × Protest × Down > Up (Pop)						
Observations	163,523,814	163,523,814	163,523,814	163,523,814	163,523,814	163,523,814
N Assembly Constituencies	808	808	808	808	808	808
Grid FE	Y	Y	Y	Y	Y	Y
Relative Time FE	Y	Y	Y	Y	Y	Y
Legislature FE	N	Y	Y	N	Y	Y
Province Trend FE	N	N	Y	N	N	Y
Mean DV	446.213	446.213	446.213	446.213	446.213	446.213
Mean DV2 (Down>Up=1)				440.214	440.214	440.214

Notes: (Columns 1-3) Results from Equation 6 of the effect of protest occurence on fires. (Columns 4-6) Results from Equation 6 with the additional interaction for pollution visibility. *Down > Up* is a dummy equal to one in months and grids from which most of the assembly constituency area is exposed to smoke (see Section 3.2 for details). The table reports the mean of the dependent variable in control grids before the cohort event-time. All regressions control for average wind speed and direction. Standard errors are clustered at the buffer area × assembly constituency level. Write why we look at linear combination as triple interaction is not significant

Figure C.3: Effect by relative time period from protest

(a) Event-Study (b) De-Trended Event-Study (c) Honest DiD



Notes: (a) Results from Equation 6. (b) Results from Equation 2 after de-trending for the pre-treatment period linear trend following **?**. (c) Sensitivity analysis for the average post-treatment effect allowing for violations of parallel trends (**?**). The reference category is $t = -1$,i.e. the month before the protest. The regressions includes grid-cohort, and province electoral cycle fixed effects, and controls for province linear time trends, average wind speed and direction. Standard errors are clustered at the buffer area × assembly constituency level. The control group includes never treated grids only.

D Adverse Selection

This section presents additional results for the adverse selection analysis. First, we present descriptive statistics for the main variables (Table D.1). Second, we present robustness checks using a de-trended event-study along with the Honest DiD sensitivity analysis following **?** (Figure D.1). Lastly, we report heterogeneity by interacting treatment with constituency-level agricultural characteristics (Figure D.2) and with constituency exposure to smoke (Table D.2).

D.1 Descriptives

Table D.1: Descriptive Statistics

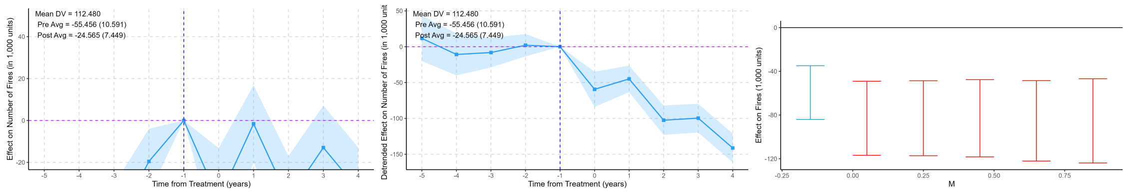
	Mean	SD	Min	Max	Observations	Unique Obs.
Grid ID					25,625,039	61,577
Year					25,625,039	11
Month					25,625,039	12
Assembly Constituency (AC)					25,625,039	689
Province					25,625,039	4
Election Year					25,625,039	9
Cohort					25,625,039	7
Legislature					25,625,039	12
Relative year	-1.595				25,625,039	21
Agricultural Politician					25,625,039	2
Number of Fires (in 1,000 units)	172.666				25,625,039	45
High Rice Production (AC level)	0.466				25,625,039	2

Notes: This table shows descriptive statistics of the main variable for the adverse selection results

D.2 Robustness

Figure D.1: Effect by relative time period from switches to an agriculturalist politician

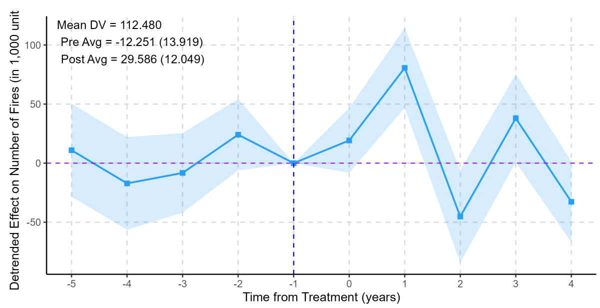
(a) Event-Study (b) De-Trended Event-Study (c) Honest DiD



Notes: (a) Results from Equation 7. (b) Results from Equation 7 after de-trending for the pre-treatment period linear trend following **?**. (c) Sensitivity analysis for the average post-treatment effect allowing for violations of parallel trends (**?**). The reference category is $t = -1$,i.e. the month before the switch to an agriculturalist. The regressions includes grid-cohort, and province electoral cycle fixed effects, and controls for province linear time trends, average wind speed and direction. Standard errors are clustered at the buffer area $\times$ assembly constituency level. The control group includes never treated grids only.

D.3 Additional Results

Figure D.2: Effect of switching to an agriculturalist politician, by constituency agricultural characteristics



Notes: Results from Equation 7 with an additional interaction for the constituency agricultural characteristics. The dummy *Above Median Rice Production* takes value 1 for constituencies with above median rice production. The regression includes grid-cohort, and legislature fixed effects, and controls for province linear time trends, average wind speed and direction. Standard errors are clustered at the grid and assembly constituency $\times$ month-year level.

Table D.2: Effect of switching to an agriculturalist politician on number of fires, by constituency exposure to smoke

	(1)	(2)	(3)	(4)	(5)	(6)
	Number of Fires (in 1,000 units)					
Post × Agriculturalist	26.660***	11.300**	11.212**	25.908***	9.939*	9.918
	(4.519)	(4.843)	(4.851)	(5.725)	(6.021)	(6.033)
Down > Up (Pop)						
Post × Down > Up (Pop)						
Agriculturalist × Down > Up (Pop)						
Post × Agriculturalist						
Observations	25,625,039	25,625,039	25,625,039	25,625,039	25,625,039	25,625,039
N Assembly Constituencies	689	689	689	689	689	689
Grid FE	Y	Y	Y	Y	Y	Y
Relative Time FE	Y	Y	Y	Y	Y	Y
Legislature FE	N	Y	Y	N	Y	Y
Province Linear Time Trend FE	N	N	Y	N	N	Y
Mean DV	120.687	120.687	120.687	120.687	120.687	120.687
Mean DV (Down > Up=1)				121.718	121.718	121.718

Notes: (Columns 1-3) Results from Equation 7 of the effect of switching to electing a politician with agricultural ties on fires. (Columns 4-6) Results from Equation 7 with the additional interaction for constituency exposure to smoke. *Down* $>$ *Up* is a dummy equal to one in months and grids from which most of the assembly constituency area is exposed to smoke (see Section 3.2 for details). The table reports the mean of the dependent variable in control grids before the cohort event-time. All regressions control for average wind speed and direction. Standard errors are clustered at the assembly constituency × election year level.

E Political Incentives Mobilize Government Machinery

Table E.1: Pollution Exposure and Politicians' Actions, all categories

Missing table: tables/action˙final˙additional˙disha

Notes: The table reports politicians' actions by level of constituency exposure. *Exposed Population Share* is the total population that upwind grids in a constituency-month would expose, normalized by the constituency's total population, and standardized (z-scored) within the estimation sample. We use DISHA documents to report the probability that a document is recorded per 100 in a given month-year, for the following categories: development (column 1), elections (column 2), governance (column 3), and welfare (column 4). The table reports the mean of the dependent variable in the sample. All regressions control for wind direction and average wind speed. Standard errors are clustered at the constituency level.